

# WHAT MATTERS FOR REPRESENTATION ALIGNMENT: GLOBAL INFORMATION OR SPATIAL STRUCTURE?

Anonymous authors  
Anonymous submission

## ABSTRACT

Representation alignment (REPA) has emerged as one of the most effective techniques for accelerating generative training, distilling the representations of a strong pretrained vision encoder into the intermediate features of a diffusion transformer. The mechanism responsible for its gains, however, has remained unresolved, and this paper settles the question directly. We ask which property of the target representation actually governs generation—its *global* semantic content (e.g., as measured by ImageNet-1K accuracy) or its spatial structure (i.e., the pairwise cosine similarity among patch tokens)—and overturn the prevailing assumption that stronger global semantic performance yields a better alignment target. Through a large-scale empirical analysis spanning 27 vision encoders and three diffusion model scales, we establish the opposite conclusion: it is spatial structure, and not global performance, that determines the generative utility of a target representation. To render this dependence quantitative, we introduce a family of fast-to-compute spatial self-similarity metrics whose values track generation quality far more closely (Pearson  $|r| > 0.85$ ) than does linear probing accuracy ( $|r| = 0.26$ ), a more than threefold gain in predictive power. Building directly on this account, we derive two minimal modifications ( $<4$  lines of code) that explicitly accentuate the transfer of *spatial* information: we replace the standard MLP projection layer in REPA with a single convolutional layer that preserves neighborhood relationships by construction, and we apply a spatial normalization layer to the external representation to sharpen its spatial contrast. The resulting method, iREPA, consistently accelerates the convergence of REPA across a broad range of vision encoders, diffusion model sizes, and training recipes—spanning REPA, REPA-E, MeanFlow, and JiT. Taken together, these results reframe the working mechanism of representation alignment and chart a concrete path toward exploiting it for markedly more efficient training of generative models.

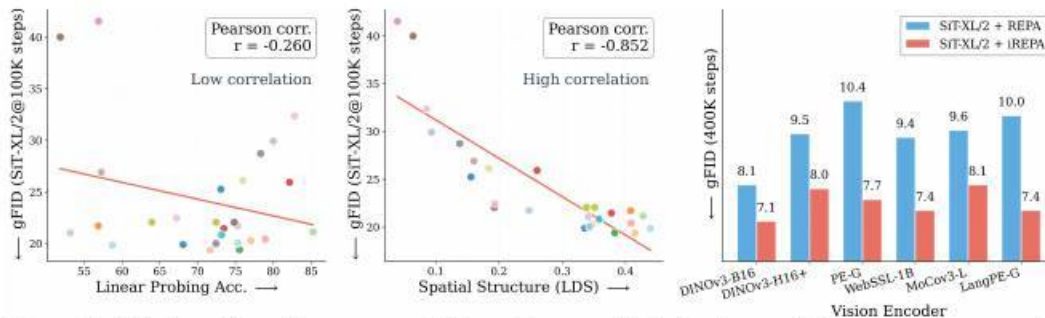


Figure 1: **What matters for representation alignment?** *Left:* A correlation analysis across 27 diverse vision encoders establishes that, contrary to the prevailing assumption, it is the spatial structure of a target representation—and not its global performance, as measured by linear probing accuracy—that governs generation performance. *Right:* Building directly on this finding, we introduce two minimal modifications that accentuate the transfer of spatial features from the target representation to the diffusion model; the resulting method, iREPA, consistently accelerates the convergence of REPA across diverse settings.



Figure 2: **Motivating examples — spatial structure matters.** These comparisons expose a clear inversion between ImageNet accuracy and generation quality. *Left:* PE-G, despite far higher validation accuracy (82.8% vs. 53.1%), yields worse generation than SpatialPE-B (Bolya et al., 2025). *Right:* WebSSL-1B (Fan et al., 2025) likewise attains much higher global performance (76.0% vs. 53.1%) yet generates worse. **Spatial Self-Similarity:** Spatial structure, by contrast, is the reliable predictor of generation quality in both comparisons; see §3 for the spatial structure metric. All results reported at 100K iterations using SiT-XL/2 and REPA.

## 1 INTRODUCTION

Representation alignment has become one of the most effective techniques for accelerating the training of diffusion transformers (Ma et al., 2024; Peebles & Xie, 2023). By aligning the internal representations of a diffusion transformer with those of a pretrained self-supervised visual encoder, recent methods deliver substantial gains in both convergence speed and final generation quality (Yu et al., 2024; Leng et al., 2025a). These empirical successes, however, remain only partially understood, because the precise mechanism through which self-supervised features enhance diffusion training has stayed an open problem. The gap raises a foundational question: does the improvement arise principally from richer global semantic information, as commonly quantified through linear probing accuracy, or from the faithful transfer of spatial structure, characterized by the pairwise relationships among patch token representations? Resolving this question is precisely what our work sets out to accomplish, and doing so reorganizes how a target representation should be chosen and exploited.

The answer carries direct practical consequence, since it dictates how practitioners should select a target representation and how they should extract its full benefit. The prevailing assumption holds that an encoder’s suitability for representation alignment tracks its ImageNet-1K validation accuracy—a proxy for global semantic understanding (Oquab et al., 2024; Chen et al., 2021). Under this view, target representations with higher ImageNet accuracy are expected to yield better generation (Yu et al., 2024). The assumption is moreover self-reinforcing: gains in the linear probing accuracy of diffusion features are routinely adduced as evidence for the effectiveness of representation alignment, further entrenching global semantic information as the presumed primary driver of improvement. The following passage from REPA (Yu et al., 2024) crystallizes this conventional understanding:

*“When a diffusion transformer is aligned with a pretrained encoder that offers more semantically meaningful representations (i.e., better linear probing results), the model not only captures better semantics but also exhibits enhanced generation performance, as reflected by improved validation accuracy with linear probing and lower FID scores.”*

In this paper, we overturn this conventional wisdom through a systematic investigation of what truly drives representation alignment: global semantic information or spatial structure. Grounding the analysis in a large-scale empirical study across a broad and diverse set of vision encoders—including recent large-scale vision foundation models such as WebSSL (Fan et al., 2025), DINOv3 (Siméoni et al., 2025), perception encoders (Bolya et al., 2025), and C-RADIO (Heinrich et al., 2024)—we arrive at three mutually reinforcing findings that together reshape this understanding and displace accuracy with a measurable property of the target representation.

**Higher validation accuracy does not imply a better representation for generation.** Contrary to the prevailing assumption, vision encoders with stronger global semantic performance—as measured by ImageNet-1K linear probing accuracy—frequently *underperform* as alignment targets, and often by a wide margin. Consider PE-Spatial-B, a spatially tuned model derived from PE-Core-G (Bolya

et al., 2025). Despite attaining far lower patch-token validation accuracy<sup>1</sup> than PE-Core-G (53.1% vs. 82.8%), PE-Spatial-B yields markedly better generation under REPA (Figure 2). WebSSL-1B (Fan et al., 2025) exhibits precisely the same inversion: substantially higher global accuracy (76.0% vs. 53.1%) coexists with worse generation. Far from anecdotal, this inversion recurs consistently across a wide range of target representations, encoder families, and controlled interventions, as we demonstrate in §2—the regularity itself signaling a governing principle behind how representation alignment benefits diffusion training.

**Spatial structure, not global information, determines generation performance.** To render this insight quantitative, we introduce a family of straightforward, fast-to-compute metrics that measure the spatial self-similarity structure (Shechtman & Irani, 2007) among patch tokens (§3)—quantifying the degree to which spatially proximate patches are more similar than distant ones. Equipped with these metrics, we conduct a large-scale correlation analysis that relates REPA generation FID to linear probing accuracy and to spatial structure across 27 vision encoders and three diffusion model scales (SiT-B, SiT-L, SiT-XL). The outcome is decisive: every spatial structure metric correlates strongly with generation FID (Pearson  $|r| > 0.852$ ), far surpassing the near-negligible predictive power of ImageNet-1K validation accuracy ( $|r| = 0.26$ ). Complementary controlled experiments corroborate the conclusion, establishing that representations carrying very little global information can nonetheless deliver substantial REPA gains: SAM2, for instance, yields better generation than encoders with roughly 60% higher ImageNet accuracy (Fig. 3), and even classical, purely spatial descriptors such as SIFT (Lowe, 1999) and HOG (Dalal & Triggs, 2005)—though not the strongest targets—produce meaningful gains under representation alignment (§3).

**Accentuating spatial features improves convergence.** Motivated by this mechanistic account, we derive two minimal modifications (<4 lines of code) that explicitly amplify the transfer of *spatial* information from the target representation to the diffusion model. First, we introduce a spatial regularization layer that sharpens the spatial contrast of the target representations by suppressing their dominant global component. Second, we identify the standard MLP projection layer—used to map diffusion features onto the target representation dimensionality—as a systematic source of spatial information loss (Figure 6a), and replace it with a single convolutional layer that preserves local spatial relationships by construction. The effect is both pronounced and consistent: the resulting method, iREPA, reliably accelerates the convergence of REPA across diverse encoders, diffusion model sizes, and training recipes, including REPA-E (Leng et al., 2025a) and MeanFlow (Geng et al., 2025) with REPA.

In summary, this paper makes the following principal contributions:

- **A mechanistic account of representation alignment.** Through a large-scale empirical analysis spanning 27 vision encoders and three diffusion model scales, we establish that it is spatial structure—not global semantic information—that drives the effectiveness of representation alignment, overturning the prevailing accuracy-centric view and replacing it with a property that is directly measurable from the target representation.
- **A predictive spatial structure metric.** We introduce a straightforward, fast-to-compute Spatial Structure Metric (SSM) that correlates substantially more strongly with downstream FID than linear probing accuracy (Pearson  $|r| > 0.85$  vs. 0.26)—a more than threefold advantage—and that furnishes a single, principled explanation for generation trends left unexplained by prior work.
- **iREPA, a minimal and general recipe.** We translate this mechanistic insight into two simple modifications that accentuate the transfer of spatial information from the target representation to the diffusion features. The resulting method delivers consistent convergence-speed improvements over REPA across variations in target representation, diffusion model architecture, and training recipe (REPA, REPA-E, MeanFlow with REPA, and JiT with REPA), establishing that the principle is recipe-agnostic rather than specific to any single setup.

## 2 MOTIVATION: GLOBAL INFORMATION MATTERS LESS

We begin with a series of motivating observations that converge on a single conclusion: the global performance of a target representation, as quantified by ImageNet-1K accuracy, does not translate

<sup>1</sup>Following REPA (Yu et al., 2024), we measure the global semantic performance of an external representation via linear probing accuracy on patch tokens, since only patch tokens are used for representation alignment.

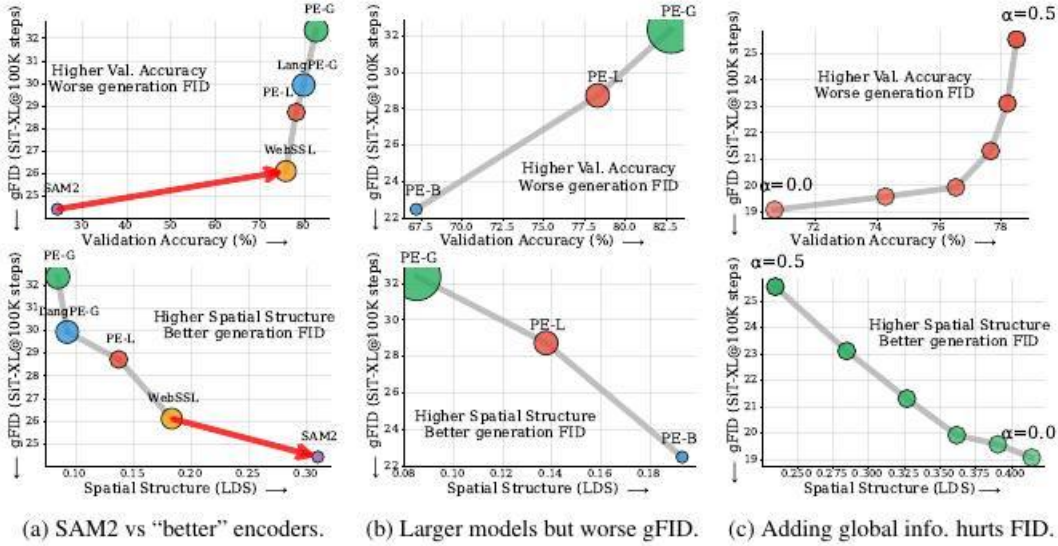


Figure 3: **Higher global information does not imply better REPA performance.** **Top row:** Across three distinct settings, global performance fails to track generation FID under REPA. (a) SAM2-S, with a validation accuracy of only 24.7%, achieves better generation with REPA than models with  $\sim 60\%$  higher validation accuracy. (b) Larger encoders within the same family can attain higher validation accuracy yet worse generation. (c) Injecting global information into patch tokens via the CLS token raises global performance but degrades generation. **Bottom row:** Spatial structure, and not global performance, is the reliable indicator of generation quality in every one of these settings; see §3 for the large-scale analysis across spatial structure metrics. All results are reported w/o classifier free guidance, SiTXL/2 w/ REPA and 250 NFE (Yu et al., 2024) for inference.

into better generation under REPA. Each observation resists explanation within the conventional, accuracy-centric view; in §3 we show that all of them are precisely and uniformly accounted for once the spatial structure of the target representation is measured.

**Recent vision encoders.** We first examine a range of recent vision encoders, including Perception Encoders (Bolya et al., 2025), WebSSL (Fan et al., 2025), and DINOv3 (Siméoni et al., 2025). Consider PE-Spatial-B (80M), a compact spatially tuned model derived from PE-Core-G (1.88B) (Bolya et al., 2025). As Figure 2 shows, PE-Core-G attains far higher ImageNet-1K accuracy (82.8% vs. 53.1%) yet performs distinctly worse as the REPA target representation (FID 32.3 vs. 21.0)—a substantial margin, and one that persists despite a more than 20-point accuracy advantage. WebSSL-1B (1.2B) exhibits the identical inversion: notwithstanding its substantially higher ImageNet-1K accuracy (76.0% vs. 53.1%), it yields worse generation as the REPA target representation (FID 26.1 vs. 21.0). In both comparisons the more accurate encoder is the weaker alignment target, in direct contradiction of the accuracy-centric assumption.

**SAM2 outperforms vision encoders with far higher ImageNet-1K accuracy.** To probe how little global information a target representation actually requires, we analyze the SAM2-S vision encoder (46M) (Ravi et al., 2024)—a small model that carries very little global information, with a validation accuracy of only 24.1% (Fig. 3a). Even so, when used for REPA, SAM2-S attains a better FID than vision encoders with dramatically higher ImageNet-1K accuracy, PE-Core-G (82.8%) among them. That a target with roughly 60 points less accuracy can nonetheless surpass far stronger encoders demonstrates that global performance is, at best, an unreliable determinant of REPA effectiveness.

**Larger models within the same encoder family can yield similar or worse generation.** It is commonly assumed that larger models within an encoder family carry better representations, as reflected in their higher ImageNet-1K accuracy. For representation alignment, however, the larger variants often produce comparable (DINOv2) or even *worse* (PE, C-RADIO) generation performance (Fig. 3b)—precisely the opposite of what the accuracy-centric view predicts. (Yu et al., 2024) report a similar observation for DINOv2, attributing it to the conjecture that “all DINOv2 models being distilled from the DINOv2-g model and thus sharing similar representations.” We later show that spatial structure supplies a more principled and general explanation of this trend—one that extends beyond DINOv2 to the PE and C-RADIO families as well (§3).

**Injecting more global information can hurt generation.** To isolate the causal effect of global information, we conduct a controlled experiment that deliberately injects global semantics—via the CLS token—into the local patch tokens (using DINOv2 as the encoder), holding all other factors fixed. The CLS-token mixing operation is  $\mathbf{p}_i^{\text{new}} = \mathbf{p}_i + \alpha \cdot \mathbf{c}$ , where  $\mathbf{p}_i$  denotes the  $i^{\text{th}}$  patch token,  $\mathbf{c}$  the CLS token, and  $\alpha \in [0, 0.5]$  modulates the mixing strength. As Figure 3c shows, increasing  $\alpha$  from 0 to 0.5 raises linear probing accuracy monotonically from 70.7% to 78.5%, while generation quality degrades in lockstep, with FID worsening from 19.2 at  $\alpha = 0$  to 25.4 at  $\alpha = 0.5$ . Injecting global information thus improves the very proxy presumed to predict REPA quality while actively harming the outcome that proxy is meant to predict—a monotonic divergence that isolates global information as a cause of degradation rather than improvement.

Taken together, these observations establish that the global performance of a representation is not a reliable indicator of its REPA effectiveness. We show next (§3) that spatial structure furnishes exactly the predictive signal that global performance lacks.

- **Finding 1.** *Stronger global semantic information (e.g., ImageNet-1K accuracy) does not imply a better representation for generation; across encoders, families, and controlled interventions, global performance fails to predict REPA effectiveness.*

### 3 SPATIAL STRUCTURE MATTERS MORE

We advance and rigorously test a single hypothesis: it is the spatial structure of a target representation, rather than its global information, that governs generation performance. To this end, we first formalize a family of straightforward metrics that quantify the spatial self-similarity structure (Shechtman & Irani, 2007) of target representations. We then establish two results that jointly confirm the hypothesis: every spatial structure metric correlates far more strongly with gFID than ImageNet-1K accuracy does, and these same metrics account, within a single unified framework, for each of the previously unexplained observations of §2.

**Measuring spatial self-similarity structure.** Given an image  $\mathcal{I}$  and vision encoder  $\mathcal{E}$ , we define:

- *Self-similarity.* Let  $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$  denote the extracted patch representations, comprising  $T = H \times W$  patch tokens. Let the kernel  $K_{\mathcal{X}}(\cdot, \cdot) \in \mathbb{R}$  measure appearance self-similarity (Shechtman & Irani, 2007) between a pair of patch tokens; throughout, we instantiate it as the cosine kernel  $K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2}$ .
- *Spatial distance.* Let  $\mathcal{P} \in \mathbb{N}^{T \times 2}$  encode the spatial location of each of the  $T$  tokens in coordinate space, and let  $d(\cdot, \cdot) \in \mathbb{N}$  denote the Manhattan distance between a pair of tokens on the resulting lattice.

We then define the *spatial self-similarity metric* as a measure of how the self-similarity  $K_{\mathcal{X}}$  varies with the lattice distance  $d$  between patch tokens. Intuitively, larger values signal stronger spatial organization, whereby spatially proximate patches are more similar to one another than distant ones. By default, we adopt a simple correlogram-contrast (local vs. distant) metric (Huang et al., 1997):

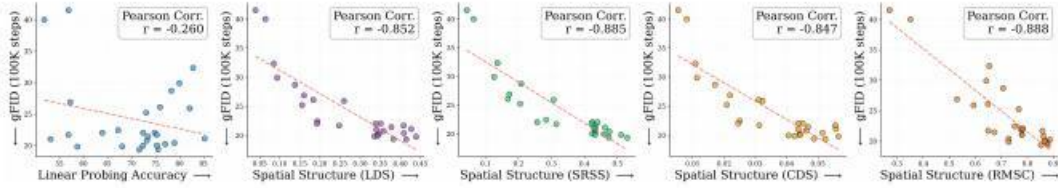
$$\text{LDS}(\mathcal{X}, \mathcal{P}) = \mathbb{E}_{d(t, t') \in (0, r_{\text{near}})} K_{\mathcal{X}}(t, t') - \mathbb{E}_{d(t, t') \geq r_{\text{far}}} K_{\mathcal{X}}(t, t').$$

The final spatial self-similarity score (LDS) is obtained as the expectation over patch representations  $\mathcal{X}$ . We fix  $r_{\text{near}} = r_{\text{far}} = H/2$  throughout; the resulting correlation is robust to their exact choice. Appendix B provides the full details together with three alternative spatial metrics (CDS, SRSS, RMSC), each of which is equally predictive (see Fig. 4), confirming that our conclusion is a property of spatial structure itself rather than an artifact of any single formulation.

**Spatial structure correlates far more strongly with generation performance than linear probing.**

We conduct a large-scale correlation analysis across 27 diverse vision encoders. As Figure 4 shows, the conventionally used linear probing accuracy exhibits only a very weak association with generation FID (Pearson  $|r| = 0.26$ ), whereas every SSM metric correlates far more strongly with generation performance (Pearson  $|r| > 0.85$ )—a more than threefold increase in predictive power that establishes spatial structure, and not global accuracy, as the reliable predictor of REPA quality.

**Generalization across diffusion model scales.** This predictive advantage is not confined to a single model size but holds uniformly across scales (SiT-B, SiT-L, SiT-XL), as verified in Figure 5. Linear



**Figure 4: Spatial structure correlates far more strongly with generation quality than linear probing.** Correlation analysis across 27 diverse vision encoders using SiT-XL/2 and REPA. Linear probing correlates only weakly with FID (Pearson  $|r| = 0.260$ ), whereas every spatial structure metric—LDS ( $|r| = 0.852$ ), SRSS ( $|r| = 0.885$ ), CDS ( $|r| = 0.847$ ), and RMSC ( $|r| = 0.888$ )—correlates far more strongly with generation performance, a more than threefold advantage that holds across all four metric definitions. See Fig. 10 for detailed plots with encoder labels.

probing remains weakly predictive at every scale ( $|r| < 0.306$ ), whereas spatial structure sustains a much stronger correlation with generation performance throughout ( $|r| > 0.826$ ), confirming that the predictive advantage of spatial structure is scale-invariant rather than an artifact of any particular model size.

- **Finding 2.** *Spatial structure correlates far more strongly with generation performance than linear probing accuracy, and the same metrics account for trends that global performance leaves unexplained.*

**Spatial structure explains previously unexplained trends.** As established in §2, global performance (ImageNet-1K accuracy) is not a predictive measure of effectiveness for representation alignment. The spatial structure metrics resolve each of those anomalies at once, supplying a single, internally consistent account of the very trends that the accuracy-centric view leaves unexplained.

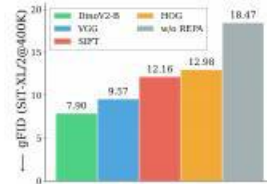
(1) *PE-Spatial-B vs. PE-Core-G.* Figure 2 shows that PE-Core-G achieves far higher ImageNet-1K accuracy (82.8% vs. 53.1%) yet performs worse as the REPA target representation (FID 32.3 vs. 22.0). Spatial structure dissolves the apparent paradox: despite its lower global performance, PE-Spatial-B exhibits substantially stronger spatial structure than PE-Core-G, and this difference directly accounts for its better generation.

(2) *SAM2 outperforms “better” vision encoders.* §2 shows that although SAM2 attains much lower ImageNet-1K accuracy (24.1%), it yields better generation than encoders with  $\sim 60\%$  higher accuracy (Fig. 3a). These gains follow directly from SAM2’s stronger spatial structure.

(3) *Larger models within the same encoder family underperform.* As shown in Fig. 3b, larger models within an encoder family attain higher ImageNet-1K accuracy yet can carry weaker spatial structure, and consequently generate worse under REPA. This mechanism moreover subsumes the observation of Yu et al. (2024) that larger models can yield similar or worse generation, replacing their distillation-based conjecture with a directly measurable property.

(4) *Adding global information to patch tokens via the CLS token hurts generation.* In Figure 3c, mixing the CLS token into patch tokens raises global performance but diminishes the spatial contrast among patch tokens, so that a token becomes highly similar to otherwise unrelated tokens (e.g., a foreground object to the background). This induced loss of spatial structure is precisely what degrades generation.

**If spatial structure matters more, can classical SIFT or HOG features serve as REPA targets?** Yes. Although they are certainly not the strongest targets, classical spatial descriptors such as SIFT (Lowe, 1999) and HOG (Dalal & Triggs, 2005), together with intermediate VGG features (Simonyan & Zisserman, 2014), all yield performance gains under REPA. That representations wholly devoid of learned global semantics can still improve generation constitutes direct evidence that representation alignment benefits from spatial structure alone, independently of any additional global information.



**Can spatial metrics explain the gains obtained with REPA?** Yes. The proposed spatial metrics account not only for the gains of standard REPA but



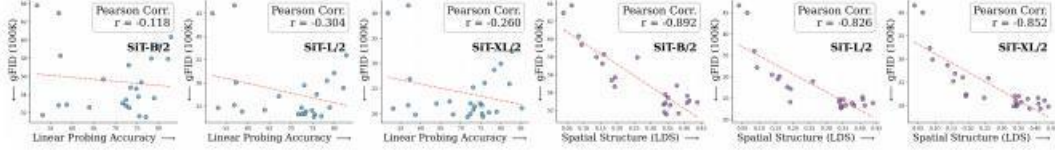


Figure 5: **Correlation analysis across model scales.** At every diffusion model scale (SiT-B, SiT-L, SiT-XL), spatial structure (right) correlates consistently more strongly with gFID than linear probing (left) does, establishing that the predictive advantage of spatial structure is scale-invariant rather than an artifact of any single model size.

also for the further improvements delivered by our enhanced training recipe (iREPA), which we introduce next in §4—so that a single spatial-structure measure explains both the baseline and its improvement.

#### 4 iREPA: IMPROVING REPRESENTATION ALIGNMENT BY ACCENTUATING WHAT MATTERS

Having established that spatial structure governs the generative utility of a target representation, we now translate this principle directly into a training recipe. We introduce two minimal, architecturally targeted modifications to the original REPA recipe, each designed to accentuate the transfer of spatial structure from the teacher (the pretrained vision encoder) to the student (the diffusion transformer). Together they constitute iREPA, a recipe that requires fewer than four lines of code yet delivers consistent and substantial improvements in convergence.

**Convolutional projection layer in place of the MLP.** Standard REPA employs a three-layer MLP projection to map the diffusion feature dimensionality onto that of the external representation. Because this projection is applied to each patch token independently, it necessarily discards the spatial relationships among tokens: as Figure 6a demonstrates, the token-wise MLP mapping is lossy and measurably attenuates the spatial contrast between patch tokens. We therefore replace it with a lightweight convolutional projection (kernel size 3, padding 1) that operates directly over the spatial grid. By coupling each token to its local neighborhood, the convolution imposes an inductive bias that preserves the neighborhood relationships among patch tokens by construction—precisely the property our analysis identifies as decisive for generation.

**Spatial normalization layer.** Consistent with Siméoni et al. (2025), the patch tokens of pretrained vision encoders carry a substantial global component, evidenced by the high linear probing accuracy obtained from the mean of the patch tokens alone (Figure 14). Figure 6b makes the underlying trade-off explicit: mixing this global component (the mean of the patch tokens) into the individual tokens raises global performance but erodes the spatial contrast between tokens, so that unrelated tokens (e.g., a foreground object and the background) come to appear spuriously similar.

The mechanistic account of §3 yields a direct prescription: deliberately suppressing this global component should sharpen the spatial contrast among patch tokens and thereby strengthen the pairwise-similarity signal that drives REPA. Acting on this prescription, we apply a simple spatial normalization layer (Ulyanov et al., 2016) to the patch tokens of the target representation:

$$\mathbf{y} = \frac{\mathbf{x} - \gamma \mathbb{E}[\mathbf{x}]}{\sqrt{\text{Var}[\mathbf{x}] + \epsilon}},$$

where  $\mathbf{x} \in \mathbb{R}^{B \times T \times D}$  denotes the patch tokens, the expectation and variance are computed across the spatial dimension,  $\gamma$  controls the fraction of the global component that is removed, and  $\epsilon = 10^{-6}$  ensures numerical stability. The single parameter  $\gamma$  thus affords direct, continuous control over the amount of global component removed, and hence over the strength of the spatial contrast supplied to the alignment objective.

##### 4.1 EXPERIMENTS

We validate iREPA through an extensive suite of experiments on ImageNet  $256 \times 256$ , structured around three research questions that together establish its effectiveness, its scalability, and the generality of the underlying spatial-transfer principle:

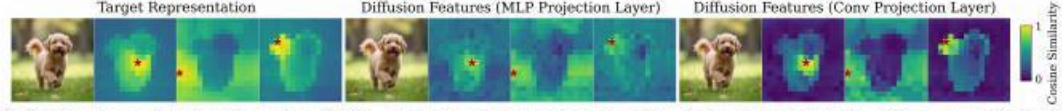
---

**Algorithm 1** Summary of key iREPA modifications to standard REPA training

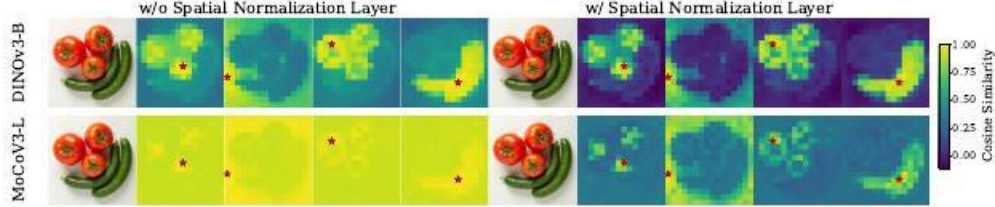
---

```
# 1. Conv projection instead of MLP
proj_layer = nn.Conv2d(D_in, D_out, kernel_size=3, padding=1)

# 2. Spatial normalization on encoder features [B, T, D]
x = x - gamma * x.mean(dim=1, keepdim=True)
x = x / (x.std(dim=1, keepdim=True) + 1e-6)
```



(a) *Simpler projection layer for REPA.* Because it acts on each patch token independently, the standard MLP projection layer in REPA (middle) discards the spatial relationships among tokens when transferring features from the target representation (left) to the diffusion features. A simpler convolutional layer, which operates directly over the spatial grid, preserves this spatial information far more faithfully (right).



(b) *Spatial normalization layer.* The patch tokens of pretrained vision encoders carry a global component that suppresses spatial contrast, causing tokens in one semantic region (e.g., tomato) to exhibit substantial cosine similarity with unrelated tokens (e.g., background or cucumber). Suppressing this global component sharpens the spatial contrast among patch tokens and thereby strengthens the pairwise-similarity signal that drives generation.



(c) *Overall impact of the improved training recipe (iREPA) on the spatial structure of the diffusion features.*

Figure 6: Motivation for the two minimal modifications (iREPA) that accentuate the transfer of spatial features from the target representation to the diffusion features. Each modification targets a distinct source of spatial information loss—the token-independent MLP projection (a) and the global overlay in the target representation (b)—and their joint effect on the diffusion features is shown in (c), where the two prove complementary. All results reported with SiTL/2 and REPA at 100K iterations.

- Does iREPA consistently accelerate the convergence of diffusion transformers over REPA across a diverse set of external representations? (Figure 7, 8, 12, 13, 15)
- Does iREPA scale with model size and generalize across variations in training configuration? (Table 1 [a,b,c], 2, 4, 5, 6, and, Figure 12, 16)
- Does iREPA generalize to more recent representation alignment methods such as REPA-E (Leng et al., 2025a) and MeanFlow w/ REPA (Geng et al., 2025), and to pixel-space diffusion models such as JiT w/ REPA (Li & He, 2025b)? (Table 3 [a,b])

**Convergence Speed.** We evaluate the convergence behavior of iREPA across a diverse set of vision encoders (DINOv3-B, WebSSL-1B, PE-Core-G, CLIP-L, MoCov3, PE-Lang-G) and two diffusion model sizes (SiT-XL/2 and SiT-B/2), reported in Fig. 7. Across every combination of target representation and model size, iREPA converges faster than baseline REPA, without a single exception—establishing that the benefit of accentuating spatial transfer is uniform rather than encoder- or scale-specific.

**Target representation.** We assess the generality of iREPA across vision encoders in Fig. 8 and Table 4. iREPA improves generation quality for every one of the vision encoders considered, with the complete comparison across all 27 encoders reported in Appendix C, E. The uniformity of the gain across this breadth of encoders confirms that what iREPA exploits is spatial transfer itself, and not any encoder-specific property.

| Enc. Size     | IS $\uparrow$ | FID $\downarrow$ | sFID $\downarrow$ | Pr. $\uparrow$ | Rc. $\uparrow$ |
|---------------|---------------|------------------|-------------------|----------------|----------------|
| PE-B (90M)    | 59.6          | 22.5             | 6.23              | 0.63           | 0.60           |
| <b>+iREPA</b> | <b>73.3</b>   | <b>17.5</b>      | <b>6.04</b>       | <b>0.66</b>    | <b>0.62</b>    |
| PE-L (320M)   | 48.6          | 28.7             | 6.53              | 0.59           | 0.60           |
| <b>+iREPA</b> | <b>76.3</b>   | <b>17.6</b>      | <b>6.36</b>       | <b>0.65</b>    | <b>0.61</b>    |
| PE-G (1.88B)  | 42.7          | 32.3             | 6.70              | 0.57           | 0.59           |
| <b>+iREPA</b> | <b>70.8</b>   | <b>19.5</b>      | <b>6.35</b>       | <b>0.64</b>    | <b>0.61</b>    |

(a) Vision Encoder size

| Model Size    | IS $\uparrow$ | FID $\downarrow$ | sFID $\downarrow$ | Pr. $\uparrow$ | Rc. $\uparrow$ |
|---------------|---------------|------------------|-------------------|----------------|----------------|
| SiT-B         | 27.5          | 49.50            | 7.00              | 0.46           | 0.59           |
| <b>+iREPA</b> | <b>34.1</b>   | <b>43.37</b>     | <b>7.87</b>       | <b>0.50</b>    | <b>0.60</b>    |
| SiT-L         | 55.7          | 24.10            | 6.25              | 0.62           | 0.60           |
| <b>+iREPA</b> | <b>66.9</b>   | <b>20.28</b>     | <b>6.14</b>       | <b>0.63</b>    | <b>0.62</b>    |
| SiT-XL        | 70.3          | 19.06            | 5.83              | 0.65           | 0.61           |
| <b>+iREPA</b> | <b>77.9</b>   | <b>16.96</b>     | <b>6.26</b>       | <b>0.66</b>    | <b>0.61</b>    |

(b) Model size

| Aln. Depth    | IS $\uparrow$ | FID $\downarrow$ | sFID $\downarrow$ | Pr. $\uparrow$ | Rc. $\uparrow$ |
|---------------|---------------|------------------|-------------------|----------------|----------------|
| Layer 4       | 28.1          | 49.0             | 6.83              | 0.46           | 0.60           |
| <b>+iREPA</b> | <b>41.7</b>   | <b>36.0</b>      | <b>6.61</b>       | <b>0.52</b>    | <b>0.62</b>    |
| Layer 6       | 26.9          | 50.6             | 7.00              | 0.46           | 0.59           |
| <b>+iREPA</b> | <b>42.4</b>   | <b>35.3</b>      | <b>6.90</b>       | <b>0.53</b>    | <b>0.62</b>    |
| Layer 8       | 24.7          | 54.8             | 7.35              | 0.44           | 0.58           |
| <b>+iREPA</b> | <b>38.3</b>   | <b>38.9</b>      | <b>7.19</b>       | <b>0.51</b>    | <b>0.62</b>    |

(c) Encoder depth (SiT-B/2)

Table 1: **Variation in training settings.** iREPA delivers consistent gains over baseline REPA across encoder size (a), diffusion model size (b), and alignment depth (c); notably, the relative gain grows with both encoder capacity and diffusion model scale—precisely the regimes in which spatial structure is most at risk, so that the trend itself corroborates the spatial-structure account. Unless otherwise specified, all results use DINOv2 as the encoder, SiT-XL/2, 100K steps, and vanilla REPA as the baseline.

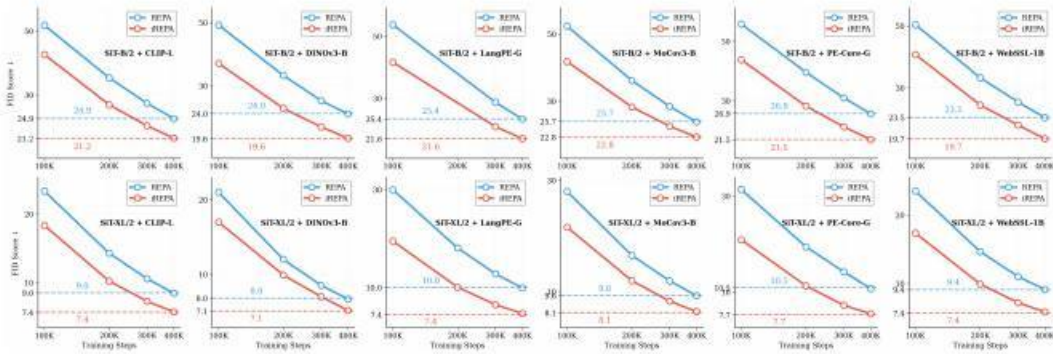


Figure 7: **Accentuating spatial features consistently improves convergence speed.** FID convergence curves show that iREPA accelerates convergence over baseline REPA across every vision encoder and diffusion model size considered, without exception.

**Encoder size.** Table 1a reports how iREPA generalizes across encoder sizes within the PE family, spanning PE-B (90M), PE-L (320M), and PE-G (1.88B). iREPA improves performance at every encoder size, and, tellingly, the relative improvement grows monotonically with encoder capacity (22.2% for PE-B, 38.8% for PE-L, and 39.6% for PE-G). This is precisely the regime in which larger encoders tend to shed spatial structure, and hence the regime in which restoring it is most valuable; the monotonic trend therefore constitutes a direct corroboration of the spatial-structure account rather than an incidental observation.

**Scalability.** Table 1b examines how iREPA scales with the size of the diffusion model. The spatial modifications improve performance at every model scale and, moreover, yield progressively larger relative gains as the model grows—establishing that the benefit of accentuating spatial transfer scales favorably with diffusion model size rather than diminishing, a property that is essential for its application to larger-scale generative training.

**Encoder depth.** Table 1c reports iREPA across alignment depths, with all results obtained on SiT-B/2 at 100K iterations using DINOv3-B as the target representation. iREPA improves consistently over baseline REPA at every alignment depth, confirming that the gains are intrinsic to the spatial modifications themselves rather than contingent on any particular choice of alignment layer.

**Ablation on individual components.** We isolate the contribution of each iREPA component in Table 2. The spatial normalization layer and the convolutional projection layer each improve generation quality over baseline REPA on their own, and the two prove complementary: applying them jointly yields the best result across every encoder. This complementarity confirms that the two modifications target distinct sources of spatial information loss—the global overlay in the target representation and the token-independent MLP mapping, respectively—and that neither subsumes the other.

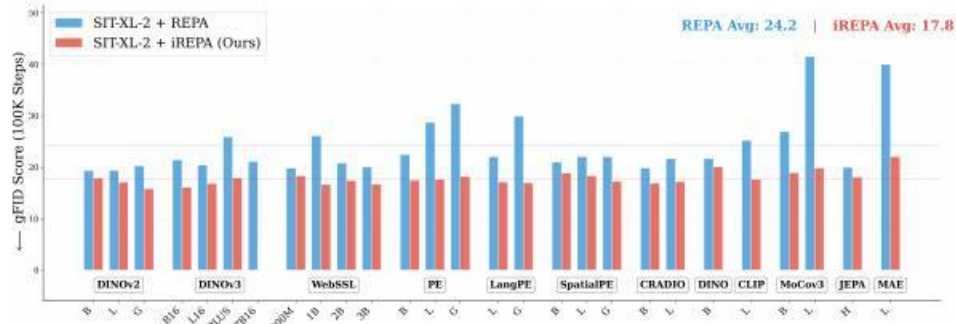


Figure 8: **Variation in target representation.** Across all 27 vision encoders, accentuating the transfer of spatial features from the target representation to the diffusion features (iREPA) yields consistent improvements in convergence speed—the uniformity confirming that the gain reflects the spatial-transfer principle rather than any encoder-specific property. See Appendix C for further results across diverse settings.

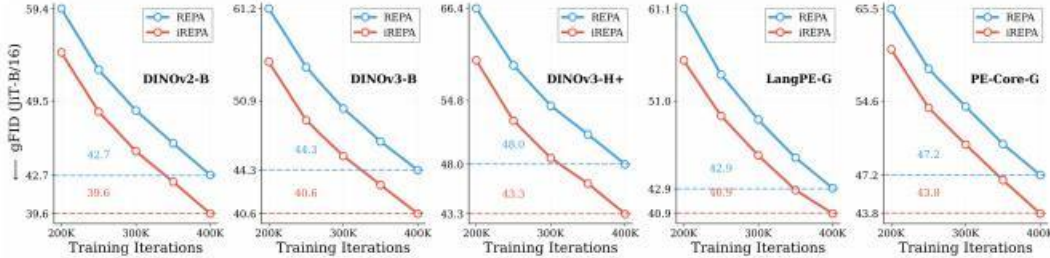


Figure 9: **Convergence comparison with pixel-space diffusion (JiT).** FID convergence curves comparing REPA and iREPA with pixel-space JiT (Li & He, 2025b) across vision encoders. Accentuating the transfer of spatial structure again yields consistently faster convergence across encoders, establishing that the principle carries over intact from latent-space to pixel-space diffusion. All results are reported with JiTB/16 (Li & He, 2025b), batch size 256, and without classifier-free guidance. See Table 10 for further results.

**Training recipe.** Finally, we test whether iREPA transfers across representation alignment recipes, evaluating it on REPA-E (Leng et al., 2025a) and MeanFlow w/ REPA (Geng et al., 2025). As Table 3 shows, the spatial modifications deliver convergence gains under both recipes, establishing that accentuating spatial transfer is a recipe-agnostic principle rather than an artifact peculiar to the original REPA setup.

**Classifier-free guidance.** Table 4 reports generation quality with classifier-free guidance across vision encoders at 400K training iterations. iREPA improves generation quality for every encoder both with and without classifier-free guidance, confirming that its gains persist under the guidance regime used at inference rather than being washed out by it.

**Pixel-space diffusion (JiT).** We further extend iREPA to pixel-space diffusion models, evaluating it on JiT-B (Li & He, 2025b). Figure 9 contrasts REPA and iREPA under JiT (Li & He, 2025b): accentuating the transfer of spatial information again yields consistently faster convergence across vision encoders (e.g., DiNOv2, DiNOv3, PE etc.), demonstrating that the principle carries over intact from latent-space to pixel-space diffusion and is therefore not specific to the latent-diffusion setting in which it was first identified.

- **Finding 3.** Accentuating the transfer of spatial structure from the target representation to the diffusion features consistently boosts convergence speed under representation alignment, across encoders, model sizes, and training recipes (REPA, REPA-E, MeanFlow, and JiT).

| Method                   | DINOv2-B     |             |             | DINOv3-B     |             |             | WebSSL-1B    |             |             | PE-Core-G    |             |             |
|--------------------------|--------------|-------------|-------------|--------------|-------------|-------------|--------------|-------------|-------------|--------------|-------------|-------------|
|                          | FID↓         | IS↑         | sFID↓       | FID↓         | IS↑         | sFID↓       | FID↓         | IS↑         | sFID↓       | FID↓         | IS↑         | sFID↓       |
| Baseline REPA            | 19.06        | 70.3        | 5.83        | 21.47        | 63.4        | 6.19        | 26.10        | 53.0        | 6.90        | 32.35        | 42.7        | 6.70        |
| iREPA (w/o spatial norm) | 18.52        | 73.3        | 6.11        | 17.76        | 74.7        | 5.81        | 21.17        | 64.6        | 6.27        | 24.97        | 57.4        | 6.21        |
| iREPA (w/o conv proj)    | 17.66        | 72.8        | 6.03        | 18.28        | 70.8        | 6.18        | 18.44        | 71.0        | 6.22        | 21.72        | 61.5        | 6.26        |
| <b>iREPA (full)</b>      | <b>16.96</b> | <b>77.9</b> | <b>6.26</b> | <b>16.26</b> | <b>78.8</b> | <b>6.14</b> | <b>16.66</b> | <b>77.5</b> | <b>6.18</b> | <b>18.19</b> | <b>75.0</b> | <b>6.03</b> |

Table 2: **Ablation on individual components.** Across every encoder, the spatial normalization layer and the convolutional projection layer (§4) each deliver substantial convergence gains over baseline REPA on their own; the two are complementary, and combining them yields the best result in every column—confirming that they address distinct sources of spatial information loss. All results reported at 100K steps with SiT-XL/2.

| Encoder         | IS↑         | FID↓        | sFID↓       | Prec.↑       | Rec.↑        |
|-----------------|-------------|-------------|-------------|--------------|--------------|
| WebSSL-1B       | 52.8        | 26.5        | 5.20        | 0.620        | 0.585        |
| <b>+iREPA-E</b> | <b>87.0</b> | <b>13.2</b> | <b>5.28</b> | <b>0.699</b> | <b>0.598</b> |
| PE-G            | 50.9        | 25.9        | 5.68        | 0.612        | 0.576        |
| <b>+iREPA-E</b> | <b>80.0</b> | <b>16.4</b> | <b>5.40</b> | <b>0.667</b> | <b>0.616</b> |
| DINOv3-B        | 82.2        | 14.4        | 4.68        | 0.694        | 0.596        |
| <b>+iREPA-E</b> | <b>93.6</b> | <b>11.7</b> | <b>4.57</b> | <b>0.703</b> | <b>0.613</b> |
| DINOv2-B        | 87.5        | 12.9        | 5.40        | 0.708        | 0.586        |
| <b>+iREPA-E</b> | <b>91.3</b> | <b>12.1</b> | <b>4.86</b> | <b>0.712</b> | <b>0.602</b> |

(a) REPA-E (SiT-XL/2)

| Encoder       | w/o CFG      |              |              |              | w/ CFG        |              |              |              |
|---------------|--------------|--------------|--------------|--------------|---------------|--------------|--------------|--------------|
|               | 4 NFE        |              | 1 NFE        |              | 4 NFE         |              | 1 NFE        |              |
|               | IS↑          | FID↓         | IS↑          | FID↓         | IS↑           | FID↓         | IS↑          | FID↓         |
| WebSSL-1B     | 27.22        | 51.40        | 24.14        | 58.67        | 87.85         | 16.59        | 69.06        | 23.74        |
| <b>+iREPA</b> | <b>31.48</b> | <b>45.67</b> | <b>27.33</b> | <b>55.70</b> | <b>100.74</b> | <b>13.89</b> | <b>78.67</b> | <b>20.69</b> |
| DINOv3-B      | 28.36        | 49.58        | 25.47        | 57.01        | 93.30         | 15.56        | 72.41        | 22.55        |
| <b>+iREPA</b> | <b>33.63</b> | <b>44.52</b> | <b>29.67</b> | <b>53.75</b> | <b>124.54</b> | <b>11.05</b> | <b>98.93</b> | <b>17.32</b> |
| DINOv2-B      | 28.77        | 48.87        | 25.43        | 56.63        | 94.40         | 15.28        | 71.95        | 22.17        |
| <b>+iREPA</b> | <b>33.63</b> | <b>44.52</b> | <b>29.67</b> | <b>53.75</b> | <b>111.44</b> | <b>12.59</b> | <b>90.25</b> | <b>18.62</b> |

(b) MeanFlow w/ REPA (SiT-B/2)

Table 3: **Variation in training recipes.** Applied on top of REPA-E (a) and MeanFlow with REPA (b), our spatial modifications yield consistent gains under both recipes, establishing that accentuating spatial transfer is a recipe-agnostic principle rather than an artifact peculiar to the original REPA setup. For MeanFlow, we report results at 1 and 4 NFEs, using a CFG value of 2.0. All results are reported at 100K training iterations.

## 5 RELATED WORK

We situate our contribution among the most directly relevant lines of work here, deferring an extended discussion to Appendix H.

**Representation alignment for generation.** A growing body of work exploits external representations to improve the training of diffusion models (Pernias et al., 2023; Fuest et al., 2024). In particular, (Yu et al., 2024; Yao & Wang, 2025; Leng et al., 2025a;b; Kouzelis et al., 2025) demonstrate that aligning the internal features of a diffusion model with the clean image features of a pretrained vision encoder yields substantial performance gains, and the idea has since been extended to video generation (Zhang et al., 2025), 3D generation (Wu et al., 2025), and the training of unified models (Ma et al., 2025a). Across these successes, however, the precise mechanism by which self-supervised features enhance diffusion training has remained unresolved. Our work closes this gap: we identify the property of the target representation that actually drives generation—its spatial structure—and translate that understanding into a concrete, improved training recipe.

**Spatial vs. global information trade-off in pretrained vision encoders.** A parallel line of work investigates the tension between global and spatial information in pretrained vision encoders (Bolya et al., 2025; Siméoni et al., 2025). (Siméoni et al., 2025) show that continued self-supervised training can increase the similarity between the global CLS token and the patch tokens, thereby degrading performance on dense spatial tasks, while (Bolya et al., 2025; Heinrich et al., 2024) explicitly train spatially tuned models for such tasks. Building on this trade-off, we establish that for generation it is the spatial structure of a vision encoder, and not its global information, that is decisive—a finding that reframes how external representations should be selected and trained for generative modeling.

## 6 CONCLUSION

This paper resolves a fundamental question about representation alignment: which property of the target representation actually drives its effectiveness—global information or spatial structure? Through a large-scale empirical analysis spanning 27 vision encoders and three diffusion model scales,

| Vision Encoder | Steps | w/o CFG       |                  |                   |                  |                 | w/ CFG        |                  |                   |                  |                 |
|----------------|-------|---------------|------------------|-------------------|------------------|-----------------|---------------|------------------|-------------------|------------------|-----------------|
|                |       | IS $\uparrow$ | FID $\downarrow$ | sFID $\downarrow$ | Prec. $\uparrow$ | Rec. $\uparrow$ | IS $\uparrow$ | FID $\downarrow$ | sFID $\downarrow$ | Prec. $\uparrow$ | Rec. $\uparrow$ |
| DINOv2-B       | 100K  | 69.20         | 19.3             | 5.89              | 0.64             | 0.61            | 157.2         | 6.35             | 5.91              | 0.769            | 0.536           |
| <b>+iREPA</b>  | 100K  | <b>77.92</b>  | <b>16.9</b>      | 6.26              | <b>0.66</b>      | <b>0.61</b>     | <b>179.3</b>  | <b>5.15</b>      | 6.23              | <b>0.783</b>     | <b>0.544</b>    |
| DINOv2-B       | 400K  | 127.4         | 7.76             | 5.06              | 0.70             | 0.66            | 263.0         | 1.98             | 4.60              | 0.799            | 0.610           |
| <b>+iREPA</b>  | 400K  | <b>128.6</b>  | <b>7.52</b>      | <b>4.89</b>       | <b>0.71</b>      | <b>0.65</b>     | <b>268.8</b>  | <b>1.93</b>      | <b>4.59</b>       | <b>0.799</b>     | <b>0.600</b>    |
| DINOv3-B       | 100K  | 63.64         | 21.4             | 6.14              | 0.63             | 0.60            | 144.0         | 7.57             | 6.09              | 0.762            | 0.526           |
| <b>+iREPA</b>  | 100K  | <b>78.79</b>  | <b>16.2</b>      | <b>6.14</b>       | <b>0.66</b>      | <b>0.61</b>     | <b>181.9</b>  | <b>4.87</b>      | 6.10              | <b>0.780</b>     | <b>0.547</b>    |
| DINOv3-B       | 400K  | 126.7         | 8.10             | 5.06              | 0.70             | 0.66            | 261.2         | 1.99             | 4.58              | 0.799            | 0.609           |
| <b>+iREPA</b>  | 400K  | <b>132.9</b>  | <b>7.13</b>      | <b>4.93</b>       | <b>0.71</b>      | <b>0.66</b>     | <b>272.4</b>  | <b>1.89</b>      | <b>4.58</b>       | <b>0.799</b>     | <b>0.600</b>    |
| WebSSL-1B      | 100K  | 53.87         | 25.5             | 6.57              | 0.61             | 0.59            | 124.0         | 9.59             | 6.37              | 0.756            | 0.515           |
| <b>+iREPA</b>  | 100K  | <b>77.47</b>  | <b>16.6</b>      | <b>6.18</b>       | <b>0.66</b>      | <b>0.61</b>     | <b>177.6</b>  | <b>5.09</b>      | <b>6.11</b>       | <b>0.787</b>     | <b>0.538</b>    |
| WebSSL-1B      | 400K  | 116.9         | 9.39             | 5.14              | 0.70             | 0.64            | 250.5         | 2.24             | 4.61              | 0.809            | 0.580           |
| <b>+iREPA</b>  | 400K  | <b>130.8</b>  | <b>7.48</b>      | <b>4.91</b>       | <b>0.70</b>      | <b>0.65</b>     | <b>271.6</b>  | <b>1.90</b>      | <b>4.58</b>       | <b>0.798</b>     | <b>0.609</b>    |
| PE-Core-G      | 100K  | 42.74         | 32.3             | 6.70              | 0.57             | 0.59            | 97.2          | 14.1             | 6.56              | 0.714            | 0.525           |
| <b>+iREPA</b>  | 100K  | <b>75.01</b>  | <b>18.1</b>      | <b>6.03</b>       | <b>0.64</b>      | <b>0.61</b>     | <b>176.8</b>  | <b>5.66</b>      | <b>6.08</b>       | <b>0.771</b>     | <b>0.544</b>    |
| PE-Core-G      | 400K  | 109.4         | 10.4             | 5.00              | 0.69             | 0.64            | 238.4         | 2.44             | 4.57              | 0.805            | 0.585           |
| <b>+iREPA</b>  | 400K  | <b>132.0</b>  | <b>7.78</b>      | 5.02              | <b>0.70</b>      | <b>0.65</b>     | <b>275.4</b>  | <b>1.93</b>      | 4.59              | <b>0.796</b>     | <b>0.606</b>    |

Table 4: **Results across encoders, with and without classifier-free guidance.** With SiT-XL/2 as the base model, iREPA improves generation quality for every encoder under both guidance settings and at both 100K and 400K steps, without a single exception. See Appendix E for detailed results on SiT-B/2 & SiT-L/2 across all encoders.

complemented by controlled interventions, we establish that it is spatial structure, and not global information, that governs the effectiveness of representation alignment, and we introduce fast-to-compute spatial metrics that render this dependence quantitatively predictive. We then operationalize the principle through two minimal modifications that accentuate the transfer of spatial information from the target representation to the diffusion features. The resulting method, iREPA, consistently accelerates the convergence of REPA across diverse vision encoders, diffusion model sizes, and training recipes (REPA, REPA-E, MeanFlow, and JiT). By reframing the working mechanism of representation alignment, these findings chart a concrete path toward selecting and training external representations far more effectively for generative modeling.

## REPRODUCIBILITY STATEMENT

We provide all implementation details and hyperparameters in Appendix G. Our code, model checkpoints, and analysis results will be released upon acceptance.

## REFERENCES

- Kristian Abstreiter, Sarthak Mittal, Stefan Bauer, Bernhard Schölkopf, and Arash Mehrjou. Diffusion-based representation learning. In *ICML 2021 Workshop on Unsupervised Reinforcement Learning*, 2021.
- Stability AI. Improved autoencoders ... <https://huggingface.co/stabilityai/sd-vae-ft-mse>, n.d. Accessed: April 11, 2025.
- Anthropic. Claude code. <https://claude.ai/code>, 2025.
- Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 15619–15629, 2023.
- Fan Bao, Shen Nie, Kaiwen Xue, Chongxuan Li, Shi Pu, Yaole Wang, Gang Yue, Yue Cao, Hang Su, and Jun Zhu. All are worth words: A vit backbone for diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 22669–22679, 2023.
- Herbert Bay, Tinne Tuytelaars, and Luc Van Gool. Surf: Speeded up robust features. In *Computer Vision—ECCV 2006: 9th European Conference on Computer Vision, Graz, Austria, May 7–13, 2006. Proceedings, Part I 9*, pp. 404–417. Springer, 2006.
- Daniel Bolya, Po-Yao Huang, Peize Sun, Jang Hyun Cho, Andrea Madotto, Chen Wei, Tengyu Ma, Jiale Zhi, Jathushan Rajasegaran, Hanoona Rasheed, Junke Wang, Marco Monteiro, Hu Xu, Shiyu Dong, Nikhila Ravi, Daniel Li, Piotr Dollár, and Christoph Feichtenhofer. Perception encoder: The best visual embeddings are not at the output of the network. *arXiv:2504.13181*, 2025.
- Shoufa Chen, Mengmeng Xu, Jiawei Ren, Yuren Cong, Sen He, Yanping Xie, Nicu Sebe, and Wai Lam. Gentron: Diffusion transformers for image and video generation. *arXiv preprint arXiv:2312.04557*, 2023.
- Shoufa Chen, Chongjian Ge, Shilong Zhang, Peize Sun, and Ping Luo. Pixelflow: Pixel-space generative models with flow. *arXiv preprint arXiv:2504.07963*, 2025.
- Xinlei Chen, Saining Xie, and Kaiming He. An empirical study of training self-supervised vision transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 9640–9649, 2021.
- Xinlei Chen, Zhuang Liu, Saining Xie, and Kaiming He. Deconstructing denoising diffusion models for self-supervised learning. *arXiv preprint arXiv:2401.14404*, 2024.
- Jang Hyun Cho, Andrea Madotto, Effrosyni Mavroudi, Triantafyllos Afouras, Tushar Nagarajan, Muhammad Maaz, Yale Song, Tengyu Ma, Shuming Hu, Hanoona Rasheed, Peize Sun, Po-Yao Huang, Daniel Bolya, Suyog Jain, Miguel Martin, Huiyu Wang, Nikhila Ravi, Shashank Jain, Temmy Stark, Shane Moon, Babak Damavandi, Vivian Lee, Andrew Westbury, Salman Khan, Philipp Krähenbühl, Piotr Dollár, Lorenzo Torresani, Kristen Grauman, and Christoph Feichtenhofer. Perceptionlm: Open-access data and models for detailed visual understanding. *arXiv:2504.13180*, 2025.
- N. Dalal and B. Triggs. Histograms of oriented gradients for human detection. In *2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR’05)*, volume 1, pp. 886–893 vol. 1, 2005. doi: 10.1109/CVPR.2005.177.
- Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pp. 248–255. Ieee, 2009.

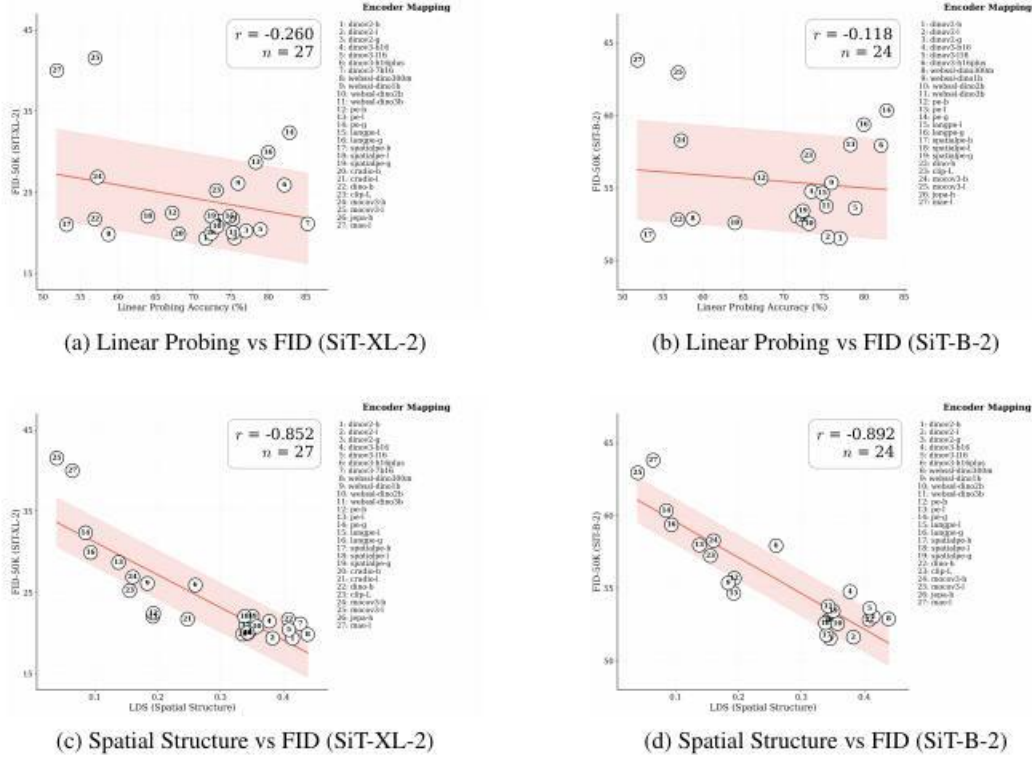
- Prafulla Dhariwal and Alexander Nichol. Diffusion models beat gans on image synthesis. *Advances in neural information processing systems*, 34:8780–8794, 2021.
- Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, Axel Sauer, Frederic Boesel, et al. Scaling rectified flow transformers for high-resolution image synthesis. In *Forty-first International Conference on Machine Learning*, 2024.
- David Fan, Shengbang Tong, Jiachen Zhu, Koustuv Sinha, Zhuang Liu, Xinlei Chen, Michael Rabbat, Nicolas Ballas, Yann LeCun, Amir Bar, et al. Scaling language-free visual representation learning. *arXiv preprint arXiv:2504.01017*, 2025.
- Michael Fuest, Pingchuan Ma, Ming Gui, Johannes Schusterbauer, Vincent Tao Hu, and Bjorn Ommer. Diffusion models and representation learning: A survey. *arXiv preprint arXiv:2407.00783*, 2024.
- Zhengyang Geng, Mingyang Deng, Xingjian Bai, J Zico Kolter, and Kaiming He. Mean flows for one-step generative modeling. *arXiv preprint arXiv:2505.13447*, 2025.
- Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial networks. *Communications of the ACM*, 63(11):139–144, 2020.
- Greg Heinrich, Mike Ranzinger, Hongxu, Yin, Yao Lu, Jan Kautz, Andrew Tao, Bryan Catanzaro, and Pavlo Molchanov. Radiov2.5: Improved baselines for agglomerative vision foundation models, 2024. URL <https://arxiv.org/abs/2412.07679>.
- Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in neural information processing systems*, 30, 2017.
- Jing Huang, S.R. Kumar, M. Mitra, Wei-Jing Zhu, and R. Zabih. Image indexing using color correlograms. In *Proceedings of IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, pp. 762–768, 1997. doi: 10.1109/CVPR.1997.609412.
- Drew A Hudson, Daniel Zoran, Mateusz Malinowski, Andrew K Lampinen, Andrew Jaegle, James L McClelland, Loic Matthey, Felix Hill, and Alexander Lerchner. Soda: Bottleneck diffusion models for representation learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 19461–19471, 2024.
- Sadeep Jayasumana, Srikumar Ramalingam, Andreas Veit, Daniel Glasner, Ayan Chakrabarti, and Sanjiv Kumar. Rethinking fid: Towards a better evaluation metric for image generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 9307–9315, June 2024.
- Dengyang Jiang, Mengmeng Wang, Liuzhuozheng Li, Lei Zhang, Haoyu Wang, Wei Wei, Guang Dai, Yanning Zhang, and Jingdong Wang. No other representation component is needed: Diffusion transformers can provide representation guidance by themselves. *arXiv preprint arXiv:2505.02831*, 2025.
- Jiatao Jing, Gu Wu, Yuanpeng Zhou, Yuxin Xu, Zixun Sun, Xinhao Ni, Yazhou Wang, Chunmeng Li, Hao Tang, Ruihua Chen, et al. Dart: Denoising autoregressive transformers for scalable text-to-image generation. *arXiv preprint arXiv:2410.08159*, 2024.
- Minguk Kang, Richard Zhang, Connelly Barnes Nga Tran, Saurabh Kar, and Jun-Yan Zhu. Distilling diffusion models into conditional gans. *arXiv preprint arXiv:2405.05967*, 2024.
- Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.
- Theodoros Kouzelis, Efstathios Karypidis, Ioannis Kakogeorgiou, Spyros Gidaris, and Nikos Komodakis. Boosting generative image modeling via joint image-feature synthesis. *arXiv preprint arXiv:2504.16064*, 2025.

- Tuomas Kynkäänniemi, Tero Karras, Samuli Laine, Jaakko Lehtinen, and Timo Aila. Improved precision and recall metric for assessing generative models. *Advances in neural information processing systems*, 32, 2019.
- Tuomas Kynkäänniemi, Tero Karras, Miika Aittala, Timo Aila, and Jaakko Lehtinen. The role of imagenet classes in fréchet inception distance. In *International Conference on Learning Representations*, 2023.
- Xingjian Leng, Jaskirat Singh, Yunzhong Hou, Zhenchang Xing, Saining Xie, and Liang Zheng. Repa-e: Unlocking vae for end-to-end tuning with latent diffusion transformers. *arXiv preprint arXiv:2504.10483*, 2025a.
- Xingjian Leng, Jaskirat Singh, Ryan Murdock, Ethan Smith, Rebecca Li, Yunzhong Hou, Zhenchang Xing, Saining Xie, and Liang Zheng. Family of End-to-End Tuned VAEs for Supercharging T2I Diffusion Transformers. <https://end2end-diffusion.github.io/repa-e-t2i/>, 2025b.
- Tianhong Li and Kaiming He. Back to basics: Let denoising generative models denoise. *arXiv preprint arXiv:2511.13720*, 2025a.
- Tianhong Li and Kaiming He. Back to basics: Let denoising generative models denoise. *arXiv preprint arXiv:2511.13720*, 2025b.
- I Loshchilov. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017.
- David G Lowe. Object recognition from local scale-invariant features. In *Proceedings of the seventh IEEE international conference on computer vision*, volume 2, pp. 1150–1157. Ieee, 1999.
- Nanye Ma, Mark Goldstein, Michael S Albergo, Nicholas M Boffi, Eric Vanden-Eijnden, and Saining Xie. Sit: Exploring flow and diffusion-based generative models with scalable interpolant transformers. In *European Conference on Computer Vision*, pp. 23–40. Springer, 2024.
- Yiyang Ma, Xingchao Liu, Xiaokang Chen, Wen Liu, Chengyue Wu, Zhiyu Wu, Zizheng Pan, Zhenda Xie, Haowei Zhang, Xingkai Yu, et al. Janusflow: Harmonizing autoregression and rectified flow for unified multimodal understanding and generation. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 7739–7751, 2025a.
- Zehong Ma, Longhui Wei, Shuai Wang, Shiliang Zhang, and Qi Tian. Deco: Frequency-decoupled pixel diffusion for end-to-end image generation. *arXiv preprint arXiv:2511.19365*, 2025b.
- Charlie Nash, Jacob Menick, Sander Dieleman, and Peter Battaglia. Generating images with sparse representations. In *International Conference on Machine Learning*, pp. 7958–7968. PMLR, 2021.
- OpenAI. Gpt-5. <https://openai.com>, 2025.
- Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *Transactions on Machine Learning Research Journal*, pp. 1–31, 2024.
- Sayak Paul. Cmmd-pytorch: Pytorch implementation of cmmd metric. <https://github.com/sayakpaul/cmmd-pytorch>, 2024. Accessed: 2025.
- William Peebles and Saining Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 4195–4205, 2023.
- Pablo Pernias, Dominic Rampas, Mats Leon Richter, Christopher Pal, and Marc Aubreville. Würstchen: An efficient architecture for large-scale text-to-image diffusion models. In *The Twelfth International Conference on Learning Representations*, 2023.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pp. 8748–8763. PMLR, 2021.

- Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Kalyan Vasudev Alwala, Nicolas Carion, Chao-Yuan Wu, Ross Girshick, Piotr Dollár, and Christoph Feichtenhofer. Sam 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024. URL <https://arxiv.org/abs/2408.00714>.
- Ethan Rublee, Vincent Rabaud, Kurt Konolige, and Gary Bradski. Orb: An efficient alternative to sift or surf. In *2011 International conference on computer vision*, pp. 2564–2571. Ieee, 2011.
- Tim Salimans, Ian Goodfellow, Wojciech Zaremba, Vicki Cheung, Alec Radford, and Xi Chen. Improved techniques for training gans. *Advances in neural information processing systems*, 29, 2016.
- Axel Sauer, Katja Schwarz, and Andreas Geiger. Projected gans converge faster. In *Advances in Neural Information Processing Systems*, volume 34, pp. 17480–17492, 2021.
- Axel Sauer, Katja Schwarz, and Andreas Geiger. Stylegan-xl: Scaling stylegan to large diverse datasets. In *ACM SIGGRAPH 2022 conference proceedings*, pp. 1–10, 2022.
- Axel Sauer, Tero Karras, Samuli Laine, Andreas Geiger, and Timo Aila. Stylegan-t: Unlocking the power of gans for fast large-scale text-to-image synthesis. *arXiv preprint arXiv:2301.09515*, 2023.
- Axel Sauer, Dominik Lorenz, Andreas Blattmann, and Robin Rombach. Adversarial diffusion distillation. In *European Conference on Computer Vision*, pp. 218–234. Springer, 2024.
- Eli Shechtman and Michal Irani. Matching local self-similarities across images and videos. In *2007 IEEE Conference on Computer Vision and Pattern Recognition*, pp. 1–8, 2007. doi: 10.1109/CVPR.2007.383198.
- Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, Francisco Massa, Daniel Haziza, Luca Wehrstedt, Jianyuan Wang, Timothée Darcet, Théo Moutakanni, Leonel Sentana, Claire Roberts, Andrea Vedaldi, Jamie Tolan, John Brandt, Camille Couprie, Julien Mairal, Hervé Jégou, Patrick Labatut, and Piotr Bojanowski. DINOv3, 2025. URL <https://arxiv.org/abs/2508.10104>.
- Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*, 2014.
- George Stein, Jesse C. Cresswell, Rasa Hosseinzadeh, Yi Sui, Brendan Leigh Ross, Valentin Villecroze, Zhaoyan Liu, Anthony L. Caterini, J. Eric T. Taylor, and Gabriel Loaiza-Ganem. Exposing flaws of generative model evaluation metrics and their unfair treatment of diffusion models. In *Advances in Neural Information Processing Systems*, 2023.
- Shitao Tang, Fuyang Chen, Aviral Joshi Bose, Kun Zhang, Shengjie Bi, Zhenyu Wang, Jiacheng Feng, Haozhe Cai, Alexander Liu, Antonio Torralba, et al. Ardit: Autoregressive diffusion transformers. *arXiv preprint arXiv:2406.00843*, 2024.
- Dmitry Ulyanov, Andrea Vedaldi, and Victor Lempitsky. Instance normalization: The missing ingredient for fast stylization. *arXiv preprint arXiv:1607.08022*, 2016.
- Runqian Wang and Kaiming He. Diffuse and disperse: Image generation with representation regularization. *arXiv preprint arXiv:2506.09027*, 2025.
- Shuai Wang, Ziteng Gao, Chenhui Zhu, Weilin Huang, and Limin Wang. Pixnerd: Pixel neural field diffusion. *arXiv preprint arXiv:2507.23268*, 2025a.
- Shuai Wang, Zhi Tian, Weilin Huang, and Limin Wang. Ddt: Decoupled diffusion transformer. *arXiv preprint arXiv:2504.05741*, 2025b.
- Ziqiao Wang, Wangbo Zhao, Yuhao Zhou, Zekai Li, Zhiyuan Liang, Mingjia Shi, Xuanlei Zhao, Pengfei Zhou, Kaipeng Zhang, Zhangyang Wang, Kai Wang, and Yang You. Repa works until it doesn't: Early-stopped, holistic alignment supercharges diffusion training. *arXiv preprint arXiv:2505.16792*, 2025c.

- 
- Haoyu Wu, Diankun Wu, Tianyu He, Junliang Guo, Yang Ye, Yueqi Duan, and Jiang Bian. Geometry forcing: Marrying video diffusion and 3d representation for consistent world modeling. *arXiv preprint arXiv:2507.07982*, 2025.
- Mengping Yang, Ceyuan Yang, Yichi Zhang, Qingyan Bai, Yujun Shen, and Bo Dai. Revisiting the evaluation of image synthesis with gans. In *Advances in Neural Information Processing Systems*, 2023.
- Jingfeng Yao and Xinggang Wang. Reconstruction vs. generation: Taming optimization dilemma in latent diffusion models. *arXiv preprint arXiv:2501.01423*, 2025.
- Sihyun Yu, Sangkyung Kwak, Huiwon Jang, Jongheon Jeong, Jonathan Huang, Jinwoo Shin, and Saining Xie. Representation alignment for generation: Training diffusion transformers is easier than you think. *arXiv preprint arXiv:2410.06940*, 2024.
- Xiangdong Zhang, Jiaqi Liao, Shaofeng Zhang, Fanqing Meng, Xiangpeng Wan, Junchi Yan, and Yu Cheng. Videorepa: Learning physics for video generation through relational alignment with foundation models. *arXiv preprint arXiv:2505.23656*, 2025.

## A DETAILED CORRELATION ANALYSIS



**Figure 10: Detailed correlation analysis with encoder labels.** The top row plots linear probing accuracy against FID for the SiT-XL-2 and SiT-B-2 models, and the bottom row plots spatial structure (LDS) against FID; each point is labeled with its encoder number (see legend). At both model scales, the spatial structure metric correlates substantially more strongly with generation quality than linear probing accuracy does, and the per-encoder labels render the tightness of the spatial-structure fit and the scatter of the linear-probing fit directly visible.

## B SPATIAL SELF-SIMILARITY METRICS

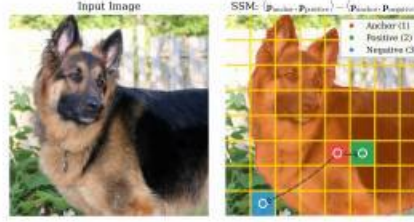


Figure 11: **Explaining the semantic-region self-similarity (SRSS) metric.** Visual explanation of the SRSS metric: for each image we compute a mask with SAM2 and sample anchor–positive–negative triplets. The SSM is the difference in cosine similarity between anchor–positive pairs (within the mask) and anchor–negative pairs (outside the mask). Intuitively, larger values indicate that patches on the same semantic region are more similar to one another than to patches on unrelated regions—a direct signature of preserved spatial structure.

**Setup.** Given an image  $\mathcal{I}$  and vision encoder  $\mathcal{E}$ , we extract patch tokens  $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$  with  $T = H \times W$  and indices  $t \in [T]$  placed on an  $H \times W$  lattice. Let  $d : [T] \times [T] \rightarrow \mathbb{N}$  be the Manhattan distance, and let the (cosine) self-similarity kernel be

$$K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2} \in [-1, 1],$$

a standard local self-similarity measure (Shechtman & Irani, 2007).

- *Patch token representation:*  $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$  where  $T = H \times W$  is the spatial grid of patches (patch index set  $[T]$ ). Also let  $d : [T] \times [T] \rightarrow \mathbb{N}$  be Manhattan distance on the  $H \times W$  grid.
- *Self-similarity kernel:*  $K : [T] \times [T] \rightarrow \mathbb{R}$  measuring self-similarity (Shechtman & Irani, 2007) between patch tokens. We use the cosine kernel  $K_{\mathcal{X}}(t, t') = \langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle / (\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2)$ .
- *Spatial self-similarity metric:* a functional  $m : \mathcal{K} \times \mathcal{D} \rightarrow \mathbb{R}$  measures how self-similarity  $K$  between patch tokens varies with lattice distance  $d$ . Intuitively, larger values indicate stronger spatial organization (near patches more similar to each other than far away patches).

We now define several straightforward metrics that quantify the spatial self-similarity structure of a target representation. Each is a low-cost functional of the self-similarity kernel and the lattice distance, and, as our correlation analysis establishes, each is far more predictive of generation quality than global accuracy—so that the finding does not rest on any single metric definition.

**Local vs. Distant Similarity (LDS).** We first adopt a simple correlogram-contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') < r_{\text{near}}] - \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') \geq r_{\text{far}}],$$

where  $r_{\text{near}}$  and  $r_{\text{far}}$  are hyperparameters, fixed by default to  $r_{\text{near}} = r_{\text{far}} = H/2$ ; the resulting correlation is robust to their exact values. A larger LDS indicates that, *on average*, spatially closer patches are more similar than distant ones—a direct signature of preserved spatial structure.

**Correlation Decay Slope (CDS).** Given the patch tokens  $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ , we compute the spatial correlogram  $g_{\mathcal{X}}(\delta) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') = \delta]$  over distances  $\delta \in \{1, \dots, \Delta\}$ , fit a least-squares line  $\hat{g}_{\mathcal{X}}(\delta) \approx \alpha + \beta\delta$  to it, and define:

$$\text{CDS}(\mathcal{X}) = -\hat{\beta}$$

where  $\hat{\beta}$  is the fitted slope. A larger CDS corresponds to a faster decay of similarity with distance, and hence to stronger spatial organization.

**Semantic-Region Self-Similarity (SRSS).** For each image, we sample a binary mask  $\mathbf{M} \in \mathbb{R}^{H \times W}$  enclosing a random object using SAM2 (Ravi et al., 2024), downsampled to  $H \times W$ . From this mask we draw three classes of points—anchor, positive, and negative—where the anchor and positive both lie within the mask (and are therefore semantically related) whereas the negative lies outside it (and is therefore less related to the anchor). An encoder that preserves spatial structure should assign

high cosine similarity to the anchor–positive pair and low similarity to the anchor–negative pair. We accordingly define the Spatial Structure Metric (SSM) as:

$$\text{SSM}(\mathbf{P}) = \mathbb{E}_{\text{anchor} \in \mathbf{M}} [\cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{pos}}) - \cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{neg}})]$$

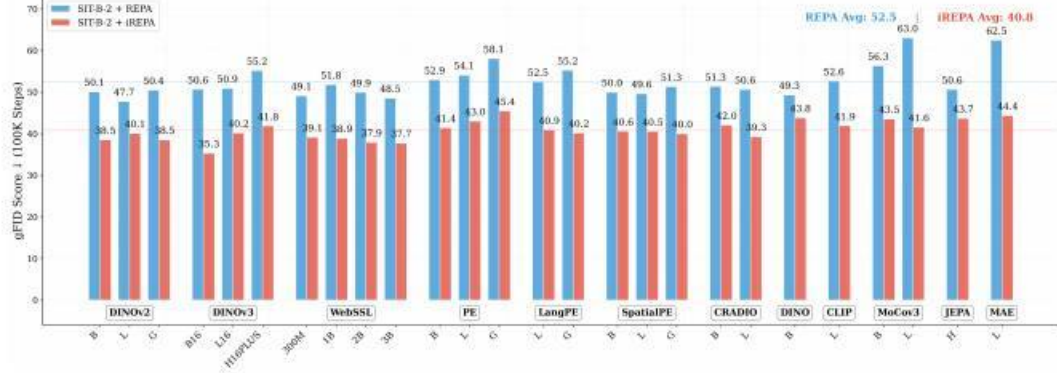
where positive points lie within Manhattan distance  $d \leq r_{\text{near}}$  of the anchor inside the mask, and negative points lie at distance  $d \geq r_{\text{far}}$  outside the mask.

**RMS Spatial Contrast (RMSC).** Finally, we introduce a simple contrast metric that captures the dispersion among patch token representations. Given  $\ell_2$ -normalized features  $\hat{\mathbf{x}}_t = \mathbf{x}_t / \|\mathbf{x}_t\|_2$  for each patch  $t$ , we compute:

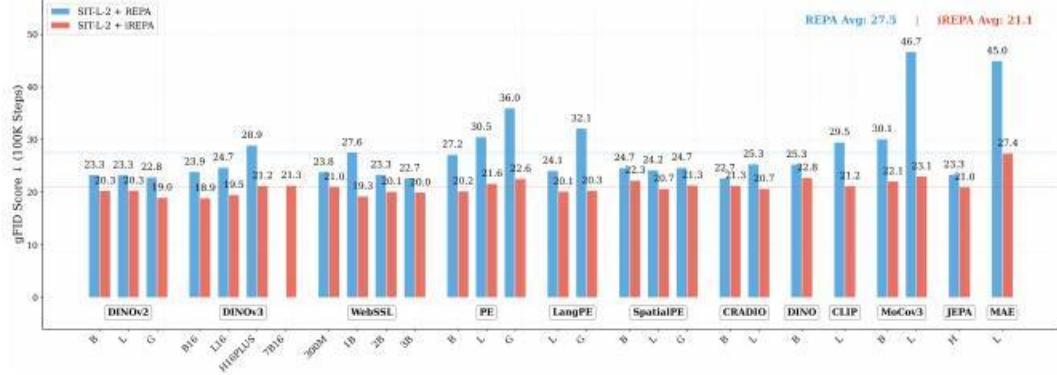
$$\text{RMSC}(\mathcal{X}) = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{x}}_t - \bar{\mathbf{x}}\|_2^2}$$

where  $\bar{\mathbf{x}} = \frac{1}{T} \sum_{t=1}^T \hat{\mathbf{x}}_t$  is the mean normalized feature across all spatial locations. A higher RMSC reflects greater spatial diversity among features, and hence preserved spatial structure; a lower RMSC reflects a more uniform feature distribution—the hallmark of the loss of spatial structure that occurs when a dominant global component overwhelms it (Siméoni et al., 2025).

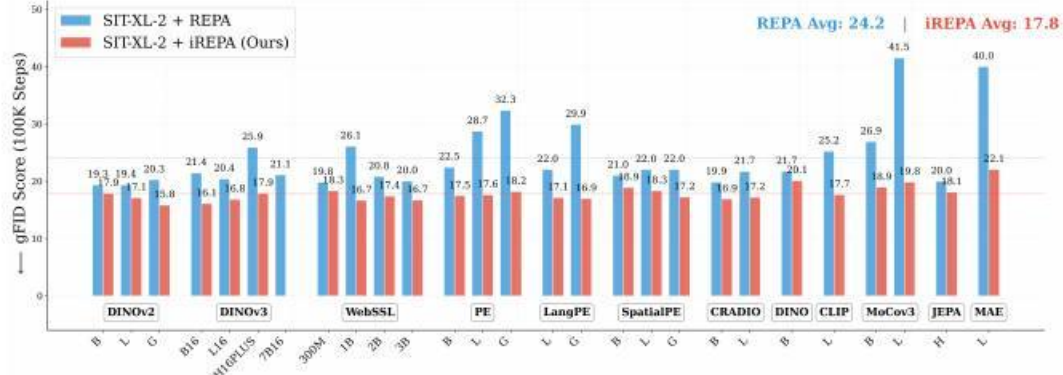
## C VISION ENCODER VARIATION



(a) SiT-B-2 (130M parameters)



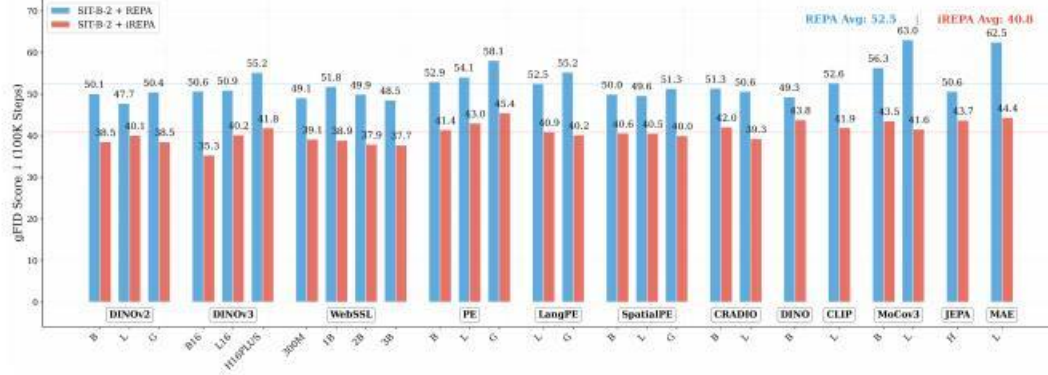
(b) SiT-L-2 (458M parameters)



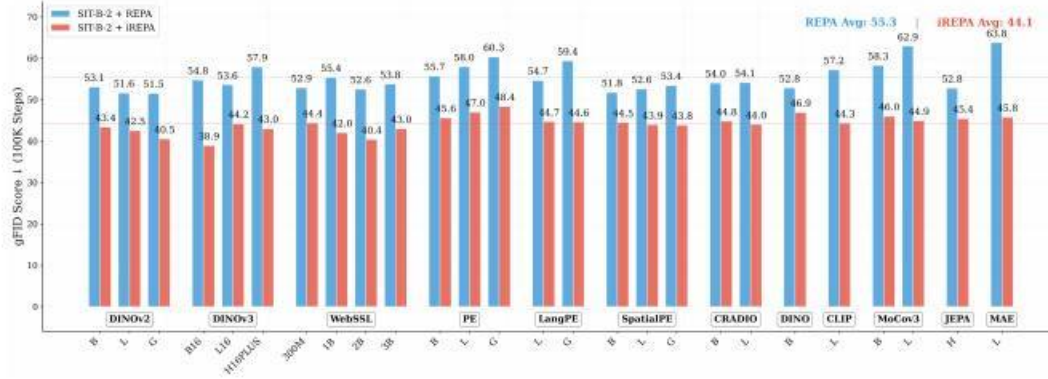
(c) SiT-XL-2 (675M parameters)

**Figure 12: Variation in vision encoders.** Ablation across vision encoders for the SiT-B-2, SiT-L-2, and SiT-XL-2 models. iREPA improves generation quality for every encoder at every model size, confirming that the benefit is uniform rather than encoder- or scale-specific.

## D ENCODER DEPTH VARIATION



(a) SiT-B-2 with 6 encoder layers



(b) SiT-B-2 with 8 encoder layers

Figure 13: **Effect of encoder depth on iREPA performance.** SiT-B-2 performance at two encoder depths (6 vs. 8 layers) across a range of vision encoders. iREPA improves consistently at both depths—slightly more so at 8 layers—and both configurations exhibit substantial FID gains once spatial regularization is applied, confirming that the benefit is intrinsic to the spatial modifications themselves rather than tied to any particular depth.

## E COMPREHENSIVE RESULTS ACROSS VISION ENCODERS

### E.1 SiT-B/2 RESULTS

| Vision Encoder | IS $\uparrow$ | FID $\downarrow$ | sFID $\downarrow$ | Prec. $\uparrow$ | Rec. $\uparrow$ |
|----------------|---------------|------------------|-------------------|------------------|-----------------|
| CLIP-ViT-L     | 59.44         | 24.84            | 6.46              | 0.584            | 0.645           |
| <b>+iREPA</b>  | <b>69.70</b>  | <b>21.26</b>     | 6.64              | <b>0.601</b>     | <b>0.644</b>    |
| DINOv2-B       | 64.92         | 22.75            | 6.54              | 0.593            | 0.653           |
| <b>+iREPA</b>  | <b>70.88</b>  | <b>21.40</b>     | 6.77              | <b>0.597</b>     | <b>0.653</b>    |
| DINOv2-L       | 66.65         | 22.46            | 6.53              | 0.594            | 0.650           |
| <b>+iREPA</b>  | <b>72.41</b>  | <b>20.99</b>     | 6.84              | <b>0.600</b>     | <b>0.650</b>    |
| DINOv3-B16     | 62.57         | 23.91            | 6.77              | 0.584            | 0.647           |
| <b>+iREPA</b>  | <b>75.48</b>  | <b>19.65</b>     | <b>6.68</b>       | <b>0.606</b>     | <b>0.649</b>    |
| DINOv3-H16+    | 61.91         | 24.82            | 6.57              | 0.579            | 0.645           |
| <b>+iREPA</b>  | <b>71.76</b>  | <b>21.80</b>     | 6.84              | <b>0.592</b>     | <b>0.655</b>    |
| DINO-B         | 58.37         | 24.71            | 6.32              | 0.586            | 0.635           |
| <b>+iREPA</b>  | <b>57.55</b>  | <b>24.41</b>     | 6.43              | <b>0.587</b>     | <b>0.638</b>    |
| LangPE-G       | 58.42         | 25.37            | 6.32              | 0.583            | 0.641           |
| <b>+iREPA</b>  | <b>66.88</b>  | <b>21.62</b>     | 6.65              | <b>0.604</b>     | <b>0.649</b>    |
| MoCov3-B       | 57.61         | 25.60            | 6.25              | 0.579            | 0.640           |
| <b>+iREPA</b>  | <b>61.41</b>  | <b>22.80</b>     | 6.46              | <b>0.596</b>     | <b>0.639</b>    |
| PE-B           | 58.86         | 25.78            | 6.44              | 0.58             | 0.64            |
| <b>+iREPA</b>  | <b>69.50</b>  | <b>21.15</b>     | 6.66              | <b>0.601</b>     | <b>0.659</b>    |
| PE-G           | 56.09         | 26.84            | 6.45              | 0.568            | 0.641           |
| <b>+iREPA</b>  | <b>70.43</b>  | <b>21.52</b>     | 6.58              | <b>0.598</b>     | <b>0.651</b>    |
| SpatialPE-B    | 59.19         | 24.28            | 6.37              | 0.586            | 0.646           |
| <b>+iREPA</b>  | <b>61.50</b>  | <b>22.71</b>     | 6.41              | <b>0.604</b>     | <b>0.644</b>    |
| SpatialPE-L    | 62.13         | 23.87            | 6.51              | 0.59             | 0.65            |
| <b>+iREPA</b>  | <b>68.83</b>  | <b>21.07</b>     | 6.59              | <b>0.607</b>     | <b>0.651</b>    |
| WebSSL-1B      | 63.20         | 23.48            | 6.33              | 0.593            | 0.646           |
| <b>+iREPA</b>  | <b>73.54</b>  | <b>19.74</b>     | 6.63              | <b>0.612</b>     | <b>0.646</b>    |
| WebSSL-2B      | 66.56         | 22.27            | 6.69              | 0.596            | 0.654           |
| <b>+iREPA</b>  | <b>72.13</b>  | <b>20.47</b>     | 6.89              | <b>0.609</b>     | <b>0.648</b>    |
| WebSSL-300M    | 62.06         | 23.64            | 6.48              | 0.590            | 0.646           |
| <b>+iREPA</b>  | <b>66.83</b>  | <b>21.46</b>     | 6.71              | <b>0.606</b>     | <b>0.653</b>    |
| WebSSL-3B      | 65.91         | 22.55            | 6.53              | 0.598            | 0.654           |
| <b>+iREPA</b>  | <b>71.98</b>  | <b>20.64</b>     | 6.71              | <b>0.607</b>     | <b>0.650</b>    |

Table 5: **SiT-B/2 performance across vision encoders at 400K iterations.** iREPA improves generation quality for every one of the vision encoders considered, delivering consistent FID and IS gains across the full set. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

## E.2 SiT-L/2 RESULTS

| Vision Encoder | IS $\uparrow$ | FID $\downarrow$ | sFID $\downarrow$ | Prec. $\uparrow$ | Rec. $\uparrow$ |
|----------------|---------------|------------------|-------------------|------------------|-----------------|
| CLIP-ViT-L     | 107.7         | 10.6             | 5.20              | 0.682            | 0.643           |
| <b>+iREPA</b>  | <b>115.8</b>  | <b>9.41</b>      | <b>5.15</b>       | <b>0.689</b>     | <b>0.653</b>    |
| DINOv2-B       | 113.3         | 9.61             | 5.11              | 0.684            | 0.653           |
| <b>+iREPA</b>  | <b>113.9</b>  | <b>9.36</b>      | <b>5.11</b>       | <b>0.690</b>     | <b>0.655</b>    |
| DINOv3-B16     | 115.3         | 9.41             | 5.24              | 0.687            | 0.651           |
| <b>+iREPA</b>  | <b>118.1</b>  | <b>9.06</b>      | <b>5.25</b>       | <b>0.685</b>     | <b>0.650</b>    |
| DINOv3-H16+    | 110.7         | 10.6             | 5.32              | 0.677            | 0.652           |
| <b>+iREPA</b>  | <b>120.5</b>  | <b>9.64</b>      | <b>5.29</b>       | <b>0.681</b>     | <b>0.663</b>    |
| LangPE-G       | 102.8         | 11.2             | 5.13              | 0.684            | 0.637           |
| <b>+iREPA</b>  | <b>112.7</b>  | <b>9.37</b>      | <b>5.02</b>       | <b>0.690</b>     | <b>0.653</b>    |
| MoCov3-B       | 102.1         | 11.0             | 5.14              | 0.685            | 0.635           |
| <b>+iREPA</b>  | <b>104.7</b>  | <b>10.1</b>      | <b>5.00</b>       | <b>0.690</b>     | <b>0.641</b>    |
| PE-G           | 98.38         | 12.6             | 5.31              | 0.670            | 0.646           |
| <b>+iREPA</b>  | <b>117.1</b>  | <b>9.65</b>      | <b>5.33</b>       | <b>0.683</b>     | <b>0.651</b>    |
| WebSSL-1B      | 107.9         | 10.5             | 5.13              | 0.686            | 0.644           |
| <b>+iREPA</b>  | <b>121.5</b>  | <b>8.62</b>      | <b>5.01</b>       | <b>0.696</b>     | <b>0.653</b>    |

Table 6: **SiT-L/2 performance across vision encoders at 400K iterations.** iREPA improves generation quality for every encoder, with especially pronounced gains in FID and IS; these results mirror the SiT-B/2 findings and confirm that the improvement holds at this intermediate model scale as well. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

## F ADDITIONAL EXPERIMENTAL RESULTS

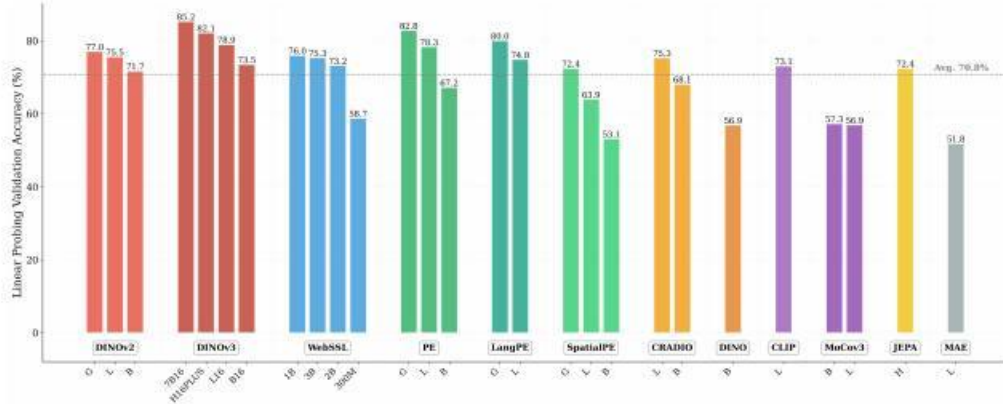


Figure 14: **Global information in mean patch tokens.** Beyond the [CLS] token, the mean of the patch tokens itself carries substantial global semantic information. While this raises global performance, it also erodes the contrast between individual patch tokens and thereby impedes spatial structure transfer. Removing part of this global component (the mean of the patch tokens) is precisely what strengthens spatial structure transfer during representation alignment (§4).

| Vision Encoder       | IS $\uparrow$ | FID $\downarrow$ | sFID $\downarrow$ | Prec. $\uparrow$ | Rec. $\uparrow$ |
|----------------------|---------------|------------------|-------------------|------------------|-----------------|
| DINOv2-B             | 262.97        | 1.98             | 4.60              | 0.799            | 0.610           |
| <b>+iREPA (Ours)</b> | <b>268.79</b> | <b>1.93</b>      | <b>4.59</b>       | <b>0.799</b>     | <b>0.600</b>    |
| DINOv3-B             | 261.18        | 1.99             | 4.58              | 0.799            | 0.609           |
| <b>+iREPA (Ours)</b> | <b>272.41</b> | <b>1.89</b>      | <b>4.58</b>       | <b>0.799</b>     | <b>0.600</b>    |
| WebSSL-1B            | 250.53        | 2.24             | 4.61              | 0.809            | 0.580           |
| <b>+iREPA (Ours)</b> | <b>271.59</b> | <b>1.90</b>      | <b>4.58</b>       | <b>0.798</b>     | <b>0.609</b>    |
| PE-G                 | 238.37        | 2.44             | 4.57              | 0.805            | 0.585           |
| <b>+iREPA (Ours)</b> | <b>275.36</b> | <b>1.93</b>      | <b>4.59</b>       | <b>0.796</b>     | <b>0.606</b>    |

Table 7: **Generation quality with classifier-free guidance.** REPA vs. iREPA with CFG (scale 2.0) across vision encoders. All experiments use SiT-XL/2 trained for 400K iterations with 250 sampling steps. iREPA improves both IS and FID for every encoder, confirming that its gains persist under the guidance regime used at inference rather than being washed out by it.

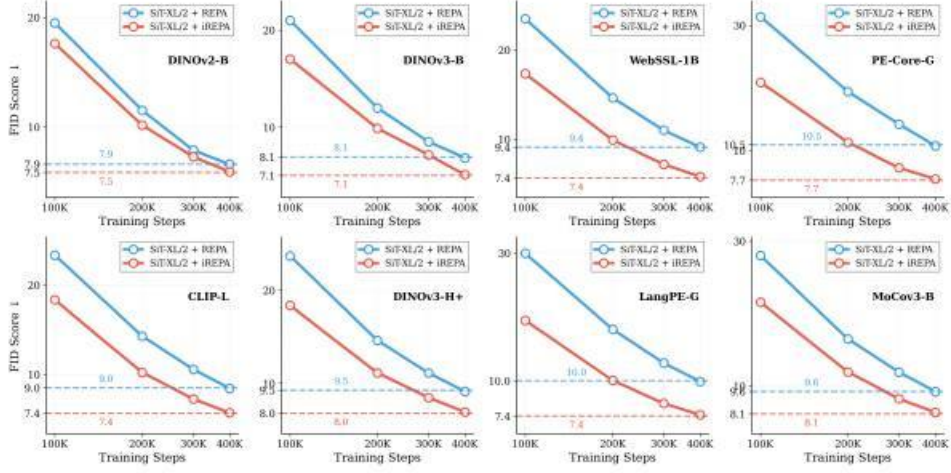


Figure 15: **Accentuating spatial features consistently improves convergence speed.** FID convergence curves for SiT-XL/2 under REPA and iREPA; iREPA converges faster than baseline REPA throughout training.

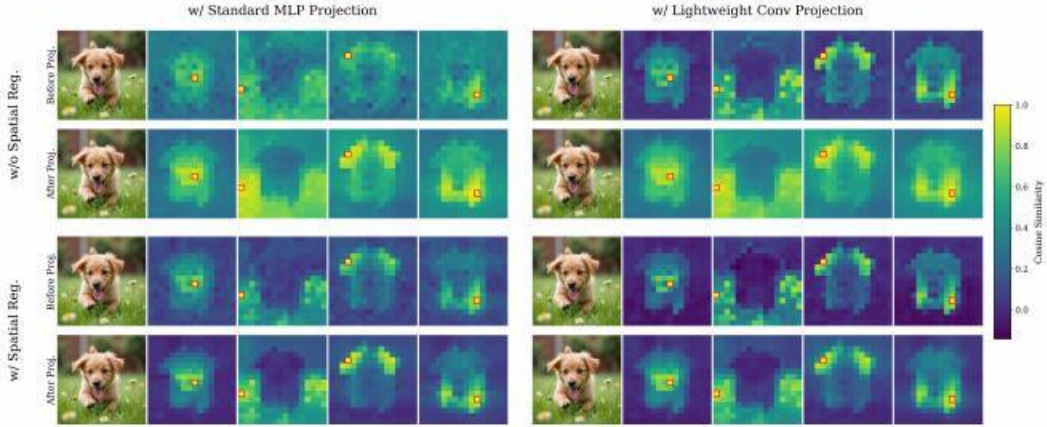


Figure 16: **Visualizing the impact of two straightforward improvements that enhance spatial feature transfer.** First, the standard MLP projection layer (top-left) discards spatial information when transferring features from the pretrained encoder (after projection) to the diffusion model (before projection). A simpler convolutional layer preserves this spatial information far more faithfully (top-right). **Second**, vision encoder features often exhibit limited spatial contrast (§2), so that a token in one semantic region (*e.g.*, dog) attains substantial cosine similarity with unrelated tokens (*e.g.*, background). We remedy this with a simple spatial regularization layer that accentuates the spatial contrast in the representation, thereby improving generation performance. The best results are obtained by combining the two modifications, which prove complementary (see Table 2).

## G IMPLEMENTATION DETAILS

**Training setup.** Unless otherwise specified, we follow the setup of REPA (Yu et al., 2024). All training is conducted on the ImageNet (Deng et al., 2009) training split. For preprocessing, we adopt the ADM (Dhariwal & Nichol, 2021) protocol—center-cropping and resizing each image to  $256 \times 256$  resolution—and we employ the `stabilityai/sd-vae-ft-mse` VAE (AI, n.d.) throughout both diffusion training and inference. The spatial normalization layer uses  $\gamma \in [0.6, 0.8]$ , and the projection layer is a convolutional layer with kernel size 3 and padding 1. Optimization uses AdamW (Kingma & Ba, 2014; Loshchilov, 2017) with a constant learning rate of  $1 \times 10^{-4}$  and a global batch size of 256. For efficiency and stability, we train with `bfloat16` mixed precision and `torch.compile`, and apply gradient clipping together with an exponential moving average (EMA) of the generative model. Aside from the two iREPA modifications, every other hyperparameter is held fixed at its REPA default, so that the reported gains are attributable to the spatial modifications alone.

For the REPA-E (Leng et al., 2025a) and JiT (Li & He, 2025a) experiments, we build directly on the official open-source implementations. For the MeanFlow (Geng et al., 2025) experiments, we obtained the implementation (including REPA) from the original authors and adapted it by incorporating the two minimal iREPA modifications (§4), leaving the remainder of the recipe unchanged.

**Evaluation.** For image generation, we follow the ADM evaluation protocol (Dhariwal & Nichol, 2021) exactly. We report generation quality using Fréchet inception distance (gFID) (Heusel et al., 2017), structural FID (sFID) (Nash et al., 2021), inception score (IS) (Salimans et al., 2016), and precision (Prec.) and recall (Rec.) (Kynkäänniemi et al., 2019), each computed over 50K generated images. For sampling, we follow REPA (Yu et al., 2024) and use the SDE Euler–Maruyama sampler with 250 steps; for JiT (Li & He, 2025a) we use 50 inference steps, matching its official implementation.

## H MORE DISCUSSION ON RELATED WORK

**Classical spatial features in computer vision.** Spatial feature extraction has long been foundational to computer vision, with classical descriptors such as SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), SURF (Bay et al., 2006), and ORB (Rublee et al., 2011) furnishing robust local representations for tasks ranging from object detection to image matching. Although these handcrafted features excel at capturing local spatial patterns and geometric invariances, recent generative modeling has concentrated almost exclusively on modern self-supervised representations selected for strong global classification performance, such as DINOv2 (Oquab et al., 2024) and CLIP (Radford et al., 2021). Our findings motivate a markedly different perspective: because spatial structure preservation is what governs generation quality, even classical spatial descriptors improve diffusion training when properly aligned. This reinstates the full spectrum of spatial feature extractors—from traditional handcrafted descriptors to modern learned representations—as viable alignment targets, provided only that they maintain strong spatial coherence.

**Pretrained visual encoders for generative models.** Pretrained visual encoders have served generative models in several capacities: as discriminators that accelerate GAN convergence (Goodfellow et al., 2020; Sauer et al., 2021; 2022; 2023; Radford et al., 2021), as teachers in adversarial distillation for diffusion models (Sauer et al., 2024; Kang et al., 2024), and, most recently, as alignment targets. In GANs (Goodfellow et al., 2020), pretrained features have both improved convergence speed and enabled scaling to large datasets, as demonstrated by StyleGAN-XL (Sauer et al., 2022) and StyleGAN-T (Sauer et al., 2023) using CLIP (Radford et al., 2021) features. For diffusion models, adversarial distillation leverages pretrained encoders to guide student networks toward higher-fidelity samples, yielding clear perceptual gains. Most directly, REPA (Yu et al., 2024) aligns diffusion features with an external encoder, establishing that representation alignment improves both convergence and generation quality. We build on this direction but relocate the emphasis from discriminative capability to spatial structure preservation—the property our analysis identifies as the decisive one.

**Denosing transformers.** Transformer architectures have become the dominant backbone for scalable generative modeling, spanning diffusion transformers (Peebles & Xie, 2023) and flow-matching variants (Ma et al., 2024). Recent architectures extend this backbone in several directions: GenTron (Chen et al., 2023) scales transformers beyond 3B parameters for text-to-image synthesis, U-ViT (Bao et al.,

2023) shows that pure transformer backbones without U-Net inductive biases remain competitive, and ARDiT (Tang et al., 2024) and DART (Jing et al., 2024) explore autoregressive formulations that couple denoising with sequential generation to trade quality against speed. Across these advances, training such models from scratch remains computationally expensive, often demanding millions of iterations to reach good generation quality. Representation alignment methods such as REPA (Yu et al., 2024) have shown that pretrained features can dramatically accelerate this convergence, yet the mechanism behind the acceleration had remained unclear. Our analysis resolves it: the benefit stems from preserving spatial structure rather than from semantic alignment—an account that explains why certain encoders accelerate training more than others and that directly informs the design of more effective alignment strategies. While (Wang et al., 2025c) propose improvements to REPA training through early stopping, they hypothesize that “REPA predominantly distills global semantic information while leaving structural information untapped.” In contrast, our analysis reveals that spatial structure (not global semantic information) already plays a very significant role in the effectiveness of REPA (§2, §3). Thus, while we indeed find that spatial structure remains underexploited (§4) in REPA, we surprisingly find that majority of improvements in REPA are already coming from introducing an inductive bias for spatial structure (not global information).

**Denoising as a self-supervised learning task.** The connection between denoising and representation learning has been examined from several angles. Early work by Abstreiter et al. (2021) extended the diffusion objective to representation learning, showing that denoising naturally acquires meaningful features; SODA (Hudson et al., 2024) couples a diffusion model with an information bottleneck to learn compact representations; and Chen et al. (2024) deconstruct diffusion models to show that a simple denoising autoencoder already attains strong self-supervised performance. Wang & He (2025) introduce a dispersive loss that encourages internal representation diversity and improves generation without external encoders, and Jiang et al. (2025) propose Self-Representation Alignment (SRA), which aligns a diffusion transformer’s latent features across noise levels to supply self-guidance without an auxiliary model. Collectively, these works establish denoising as a fundamental self-supervised task that naturally encourages robust features. Our findings sharpen this view considerably: aligning diffusion models with the spatial representations of self-supervised encoders improves both generative and discriminative capability, and pinpoints spatial structure preservation as the single factor underlying the synergy.

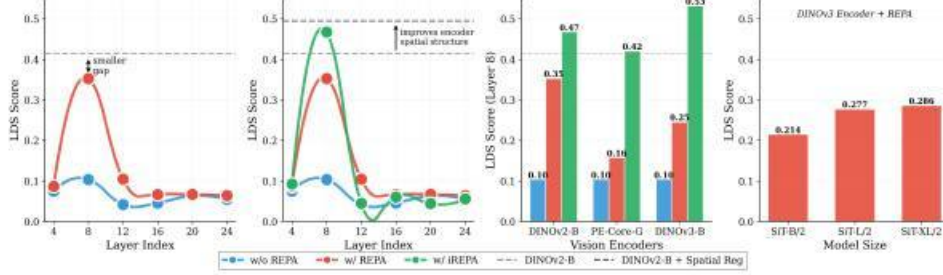
**Scaling self-supervised vision encoders.** Recent years have brought remarkable progress in scaling self-supervised vision models to unprecedented sizes and datasets. DINOv3 (Siméoni et al., 2025) trains a 7B-parameter ViT on 1.7 billion unlabeled images by aligning representations across augmentations; WebSSL (Fan et al., 2025) shows that models trained on more than 2 billion images match language-supervised models such as CLIP (Radford et al., 2021) on vision-language tasks without any language supervision; and C-RADIO (Heinrich et al., 2024) distills multiple teachers into versatile encoders that excel across visual domains. I-JEPA (Assran et al., 2023) learns by predicting masked regions in representation space rather than reconstructing pixels, SAM (Ravi et al., 2024) specializes in promptable segmentation via large-scale supervision, and the Perception Encoder family (Bolya et al., 2025; Cho et al., 2025) shows that intermediate features often surpass final representations for dense prediction. These models are almost invariably evaluated on global tasks such as image classification, yet our work exposes a crucial gap: strong discriminative performance does not translate into better generation. It is instead the encoders that preserve spatial structure—irrespective of their classification accuracy—that confer the greatest benefit for diffusion training. This argues for reconsidering the evaluation criteria of self-supervised encoders in generative applications, treating spatial coherence as no less important than semantic understanding.

## I NOTE ON LLM USAGE

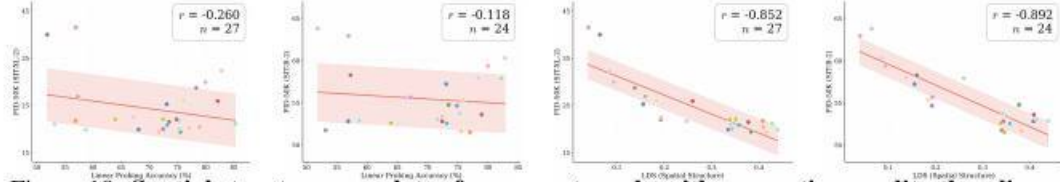
We use GPT-5 (OpenAI, 2025) to explore variations of the spatial structure metrics discussed in §3. All figures in the paper are generated directly from our experiment logs and checkpoints using Claude-Code (Anthropic, 2025). We additionally rely on LLM assistance for surveying and formulating the related work in Appendix H, and use Cursor in places to assist with paper writing.

## J ADDITIONAL DISCUSSION ON SPATIAL STRUCTURE METRICS

**Spatial structure metrics can measure the representation gap.** Yu et al. (2024) measure the representation gap through linear probing, invoking the resulting increase in validation accuracy to explain the effectiveness of REPA. We show that spatial structure metrics (SSM) capture this gap equally well while additionally accounting for improvements across models: as Figure 17 demonstrates, representation alignment closes the gap between the SSM of a pretrained encoder such as DINOv2 and that of the diffusion features—recasting the representation gap in the very spatial terms that our analysis shows to be decisive.



**Figure 17: Representation alignment as bridging the spatial feature gap.** **a)** Representation alignment can itself be read as bridging the spatial feature gap between the diffusion and vision-encoder patch features. **b)** iREPA (§4) accentuates the spatial features of the vision encoder—at the cost of some global information—thereby achieving better generation performance. **c)** iREPA improves performance consistently across vision encoders. **d)** The spatial-structure improvements scale with diffusion model size. All results are reported with SiT-XL/2 at 400k iterations.



**Figure 18: Spatial structure correlates far more strongly with generation quality than linear probing.** Correlation analysis across 27 vision encoders. **Left two:** Linear probing accuracy vs. FID for SiT-XL-2 (Pearson  $r = -0.26$ ) and SiT-B-2 ( $r = -0.12$ ) is only weakly correlated. **Right two:** Spatial structure (LDS) vs. FID for SiT-XL-2 ( $r = -0.85$ ) and SiT-B-2 ( $r = -0.89$ ) is strongly correlated at both scales. See Figure 10 in the Appendix for detailed plots with encoder labels.

## K ADDITIONAL ANECDOTAL COMPARISONS

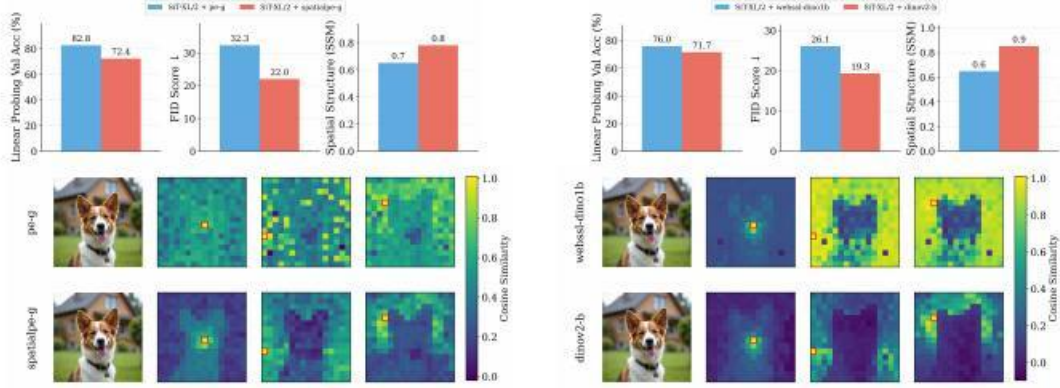


Figure 19: **Motivating examples from recent SSL encoders — spatial structure matters.** **Top row:** Metrics comparison exposing an inverse relationship between ImageNet accuracy and generation quality. **Left:** PE-g achieves higher accuracy (82.8% vs 72.4%) yet worse FID (32.3 vs 22.0) than Spatial-PE-g, together with much lower spatial structure (LDS: 0.1 vs 0.4). **Right:** WebSSL-dino1b attains higher accuracy (76.0% vs 71.7%) yet worse FID (26.1 vs 19.3) than DINOv2-b, together with weaker spatial structure (LDS: 0.2 vs 0.4). **Bottom row:** Spatial cosine similarity visualizations confirm that the encoders with better generation (Spatial-PE-g, DINOv2-b) preserve clear spatial coherence, whereas those optimized for classification (PE-g, WebSSL) forfeit spatial structure.

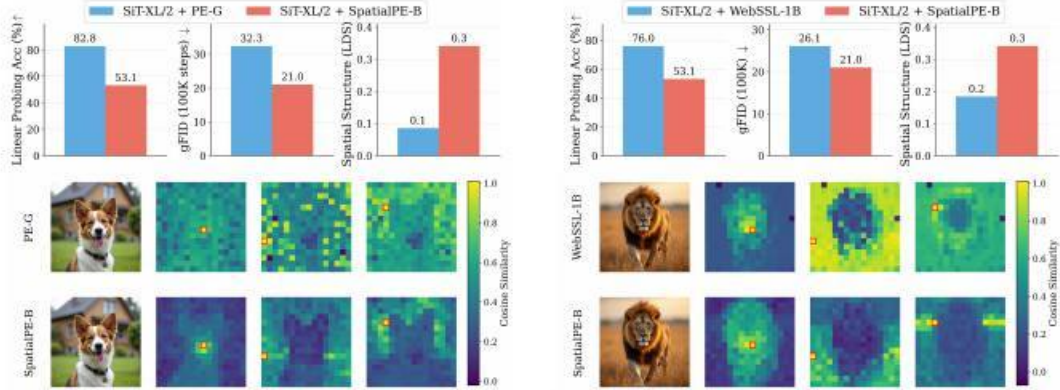


Figure 20: **Motivating examples — spatial structure matters.** **Top:** Metrics comparison exposing an inverse relationship between ImageNet accuracy and generation quality. **Left:** PE<sub>core</sub>-G, despite its significantly higher validation accuracy (82.8% vs. 53.1%), yields worse generation quality than PE<sub>Spatial</sub>-B (Bolya et al., 2025). **Right:** Likewise, WebSSL-1B (Fan et al., 2025) attains much better global performance (76.0% vs. 53.1%) yet generates worse. **Bottom:** Spatial structure, by contrast, is the reliable predictor of generation quality; see §3 for the spatial structure metric. All results reported at 100K using SiT-XL/2 and REPA.

## L FURTHER DISCUSSION

**Role of Spatial Normalization.** We visualize the effect of spatial normalization on vision encoder representations. Pretrained encoders frequently carry a substantial global component that suppresses the contrast among spatial tokens, so that tokens in one semantic region become highly correlated with tokens in unrelated regions (e.g., background) and forfeit their spatial distinctiveness. Spatial normalization removes this global overlay and sharpens the contrast between spatial tokens; the representation thereby retains its local spatial structure while shedding the global information that would otherwise interfere with generation quality.

The following figures illustrate this effect across several examples, showing how spatial normalization reshapes the feature representations so as to emphasize spatial structure:

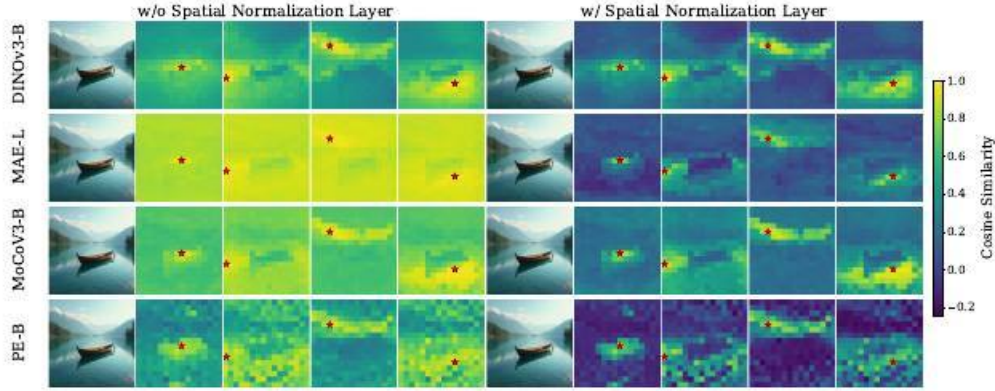


Figure 21: **Visualizing the impact of spatial normalization** (Example 1). The heatmaps depict token similarity patterns before and after spatial normalization. Without normalization (left), the dominant global component induces high correlations across unrelated regions; with spatial normalization (right), the local spatial structure is sharpened and the global interference suppressed, yielding markedly more distinct semantic boundaries.

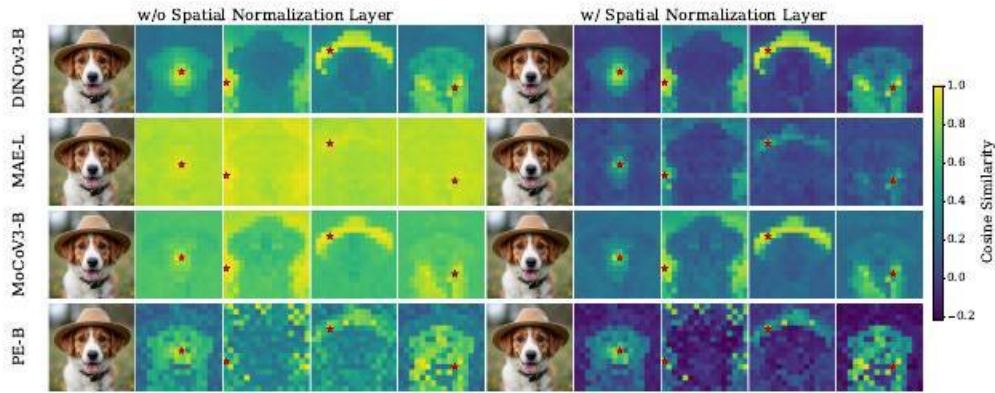


Figure 22: **Visualizing the impact of spatial normalization** (Example 3). The heatmaps depict token similarity patterns before and after spatial normalization. Without normalization (left), the dominant global component induces high correlations across unrelated regions; with spatial normalization (right), the local spatial structure is sharpened and the global interference suppressed, yielding markedly more distinct semantic boundaries.

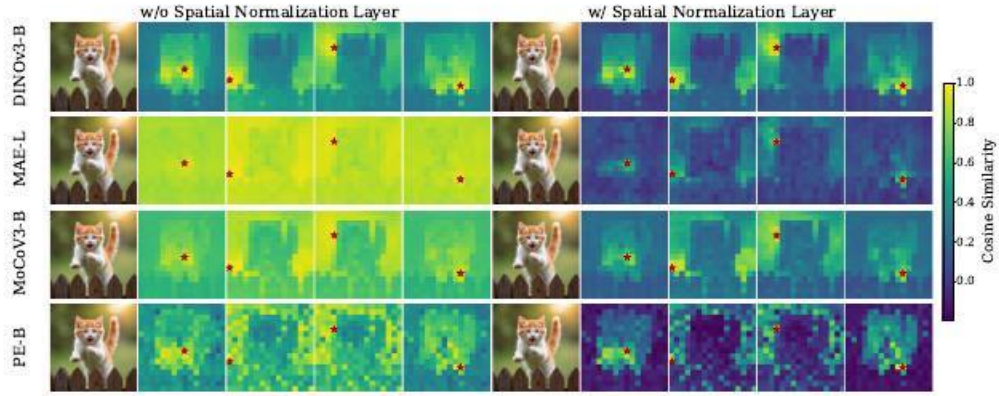


Figure 23: **Visualizing the impact of spatial normalization** (Example 2). The feature similarity maps demonstrate how spatial normalization improves spatial contrast. The original features (left) carry a global overlay that erodes the distinction between foreground and background regions; after normalization (right), the spatial tokens become more locally coherent, with a clearer separation between distinct semantic regions.

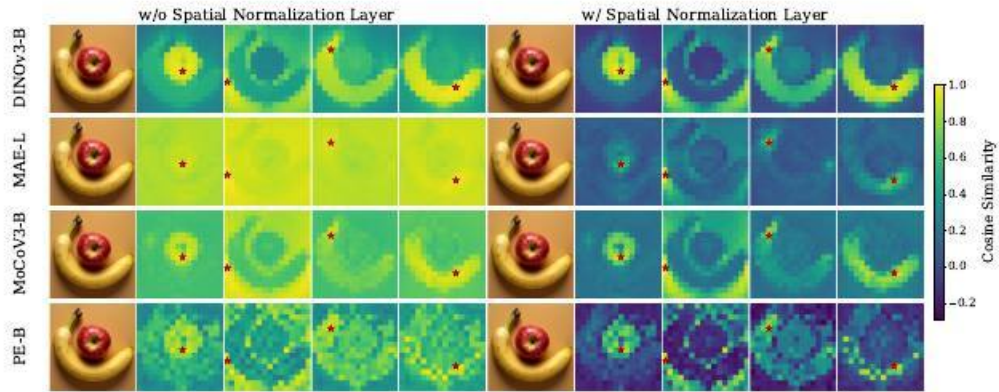


Figure 24: **Visualizing the impact of spatial normalization** (Example 4). The heatmaps depict token similarity patterns before and after spatial normalization. Without normalization (left), the dominant global component induces high correlations across unrelated regions; with spatial normalization (right), the local spatial structure is sharpened and the global interference suppressed, yielding markedly more distinct semantic boundaries.

## M ADDITIONAL RESULTS

### M.1 ADDITIONAL RESULTS AT HIGHER RESOLUTIONS

**Setup.** We conduct further experiments on ImageNet-512 (Deng et al., 2009) to assess how the proposed spatial improvements generalize to higher resolutions. Following the REPA (Yu et al., 2024) setup, we report results across a range of pretrained encoders (DINOv2, DINOv3, WebSSL, PE, etc.), using SiT-XL and SiT-B with 50 NFE at inference and without classifier-free guidance. Fig. 25 contrasts REPA before and after the spatial improvements (iREPA). The spatial improvements transfer cleanly to the higher resolution, delivering consistent gains across every pretrained encoder considered (DINOv2, DINOv3, WebSSL, PE, etc.) and thereby confirming that the principle is not tied to the  $256 \times 256$  setting in which it was first established.

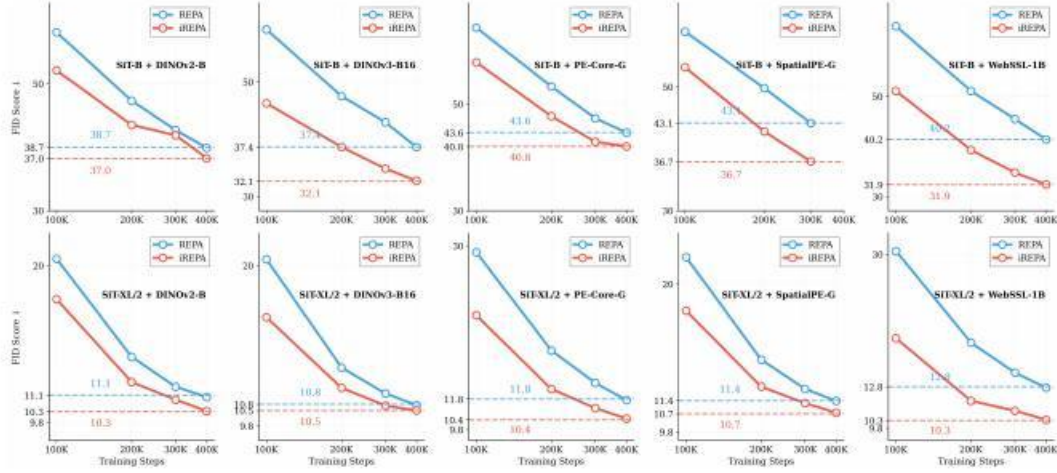


Figure 25: **Convergence results at 512x512 resolution.** Accentuating spatial features consistently improves convergence speed across resolutions, holding for both ImageNet-256 (Figure 7) and ImageNet-512 (above)—direct evidence that the benefit is resolution-invariant rather than confined to the  $256 \times 256$  setting.

## M.2 ADDITIONAL RESULTS ON MULTIMODAL T2I TASKS

**Setup.** To probe the generalizability of the proposed spatial improvements beyond ImageNet, we further conduct extensive experiments on multimodal text-to-image (T2I) tasks. Following REPA (Yu et al., 2024), we adopt MMDiT (Esser et al., 2024) as the diffusion backbone and apply REPA with a range of pretrained vision encoders (DINOv2, CLIP, WebSSL, PE, etc.). As in REPA (Yu et al., 2024), all models are trained for 150K steps with a batch size of 256 and evaluated with an ODE sampler at 50 NFE. Fig. 26 contrasts REPA before and after the spatial improvements (iREPA). The spatial improvements extend to the multimodal T2I setting as well, yielding consistent gains across every pretrained encoder considered (DINOv2, CLIP, WebSSL, PE, etc.)—demonstrating that the principle reaches beyond class-conditional ImageNet generation to conditional multimodal synthesis.

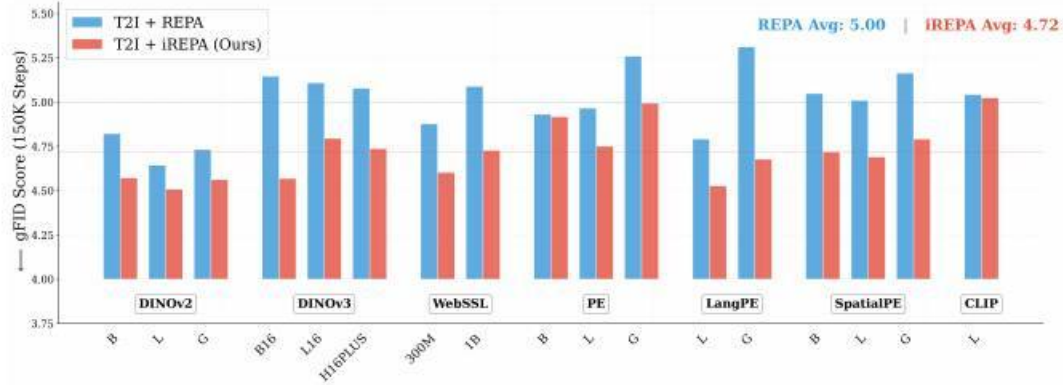
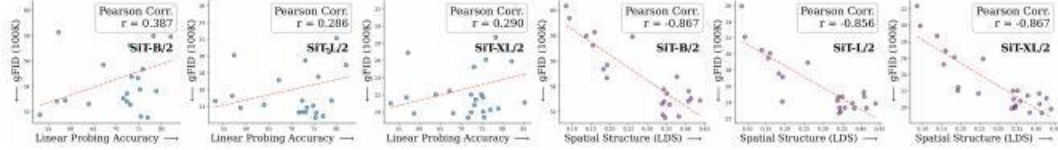


Figure 26: **Text-to-image generation across encoder variants.** Accentuating the transfer of spatial features from the target representation to the diffusion features consistently improves convergence speed for both ImageNet (see §4) and multimodal T2I tasks (above). The gains hold uniformly across the choice of pretrained encoder (DINOv2, CLIP, WebSSL, PE, etc.), demonstrating that the principle extends from class-conditional generation to conditional multimodal synthesis.

### M.3 CORRELATION ANALYSIS WITHOUT OUTLIERS (MoCOv3-L AND MAE-L)

We repeat the correlation analysis from Section 3 after removing MoCOv3-L and MAE-L, which were identified as outliers. Figure 28 reports the updated correlations across model sizes. Spatial structure continues to correlate far more strongly with generation performance than linear probing accuracy does. Strikingly, once the outliers are removed, linear probing exhibits a small *positive* correlation with gFID—that is, higher linear probing accuracy is associated with *worse* generation. This trend directly reinforces the observations of Sec. 2, in which target representations with higher global semantic performance (linear probing accuracy) frequently yield similar or worse generation under representation alignment (REPA).



**Figure 27: Correlation analysis without outliers.** After removing the outliers (MoCOv3-L and MAE-L), spatial structure continues to correlate far more strongly (Pearson  $|r| > 0.85$ ) with generation performance than linear probing accuracy (Pearson  $|r| < 0.38$ ) across all model sizes (B, L, XL). Strikingly, once the outliers are removed, linear probing exhibits a small positive correlation with gFID—higher linear probing accuracy is associated with worse generation. This trend directly reinforces the observations of Sec. 2, in which target representations with higher global semantic performance (linear probing accuracy) frequently yield similar or worse generation under representation alignment (REPA).

#### M.4 ADDITIONAL QUALITATIVE RESULTS

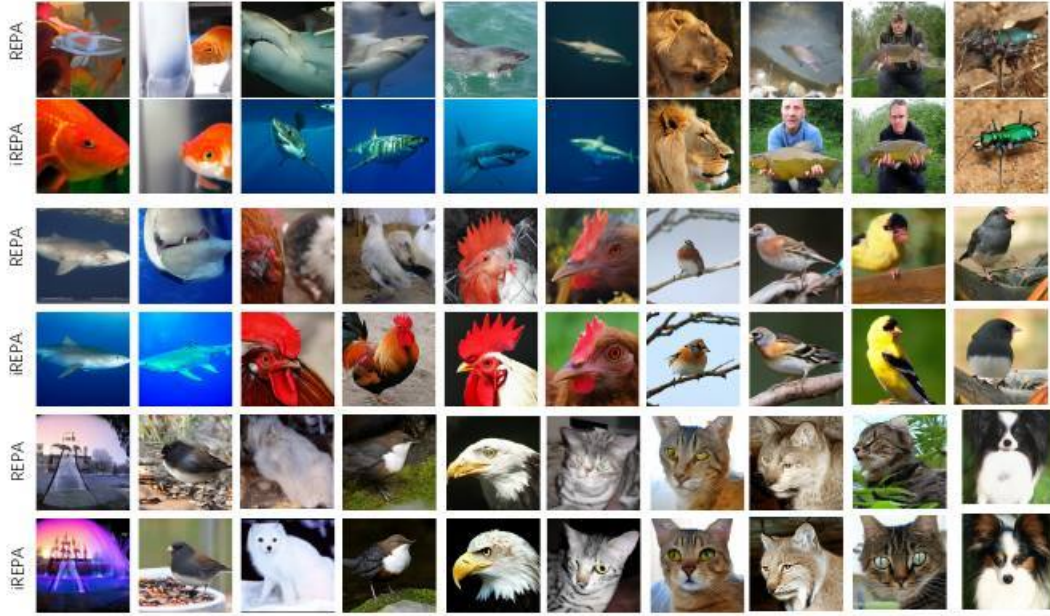


Figure 28: **Additional Qualitative Results** comparing generation outputs before and after the spatial improvements with REPA. All results use PE-G (Bolya et al., 2025) as the pretrained vision encoder, 400K steps (80 epochs), and a classifier-free guidance scale of 4.0. Mirroring the quantitative improvements (§4), the spatial improvements (iREPA) visibly enhance both the visual quality and the spatial coherence of the generated outputs.

## M.5 ADDITIONAL RESULTS WITH ALTERNATIVE EVALUATION METRICS

Beyond the traditional evaluation metrics (Inception Score, FID, sFID, Precision, Recall), we further verify the robustness of the proposed findings using alternative measures of generation quality (Stein et al., 2023; Yang et al., 2023; Kynkäänniemi et al., 2023; Jayasumana et al., 2024). Specifically, we adopt the CMMD metric (Jayasumana et al., 2024) with the PyTorch implementation of (Paul, 2024), using the standard reference set from the OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021) as reference images. We then re-verify the key findings of both §3 and §4 under CMMD.

### Spatial structure correlates far more strongly with generation performance than linear probing.

To assess the robustness of the analysis in §3, we repeat the large-scale correlation analysis across vision encoders under the CMMD metric (Jayasumana et al., 2024). Results are reported in Figure 29, using SiT-B/2 (100K steps) and REPA. Every spatial metric correlates far more strongly (Pearson  $|r| > 0.88$ ) with generation performance (CMMD) than linear probing does (Pearson  $|r| = 0.074$ ), establishing that the central findings of §3 hold irrespective of whether generation quality is measured by FID or by CMMD.

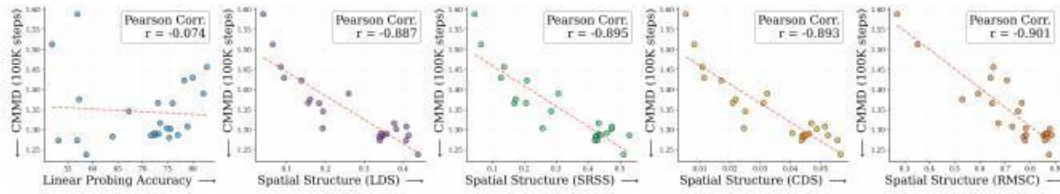


Figure 29: **Correlation analysis with the CMMD metric.** We repeat the correlation analysis of §3 with the CMMD metric (Jayasumana et al., 2024) in place of gFID. All results use SiT-B/2 (100K steps) and REPA. Every spatial metric correlates far more strongly (Pearson  $|r| > 0.88$ ) with generation performance (CMMD) than linear probing does (Pearson  $|r| = 0.074$ ), confirming that the key findings of §3 are robust to the choice of evaluation metric rather than specific to FID.

**Accentuating the transfer of spatial features improves generation quality.** We next examine the generalizability of the analysis in §4 under the CMMD (Jayasumana et al., 2024) metric, alongside the traditional evaluation metrics (gFID, sFID, IS, etc.). Results are reported in Table 8. Across the full range of vision encoders, accentuating the transfer of spatial features (iREPA) improves convergence speed under both CMMD and the traditional metrics (IS, FID, sFID, Prec., Recall), and this improvement holds consistently both with and without classifier-free guidance (CFG).

| Vision Encoder | Steps | w/o CFG      |              |             |             |             |             | w/ CFG       |              |             |             |             |             |
|----------------|-------|--------------|--------------|-------------|-------------|-------------|-------------|--------------|--------------|-------------|-------------|-------------|-------------|
|                |       | CMMD↓        | IS↑          | FID↓        | sFID↓       | Prec.↑      | Rec.↑       | CMMD↓        | IS↑          | FID↓        | sFID↓       | Prec.↑      | Rec.↑       |
| DINOv2-B       | 100K  | 0.702        | 69.20        | 19.3        | 5.89        | 0.64        | 0.61        | 0.529        | 157.2        | 6.35        | 5.91        | 0.77        | 0.54        |
| <b>+iREPA</b>  | 100K  | <b>0.652</b> | <b>77.92</b> | <b>16.9</b> | 6.26        | <b>0.66</b> | <b>0.61</b> | <b>0.484</b> | <b>179.3</b> | <b>5.15</b> | 6.23        | <b>0.78</b> | <b>0.54</b> |
| DINOv2-B       | 400K  | 0.455        | 127.4        | 7.76        | 5.06        | 0.70        | 0.66        | 0.320        | 263.0        | 1.98        | 4.60        | 0.80        | 0.61        |
| <b>+iREPA</b>  | 400K  | <b>0.438</b> | <b>128.6</b> | <b>7.52</b> | <b>4.89</b> | <b>0.71</b> | <b>0.65</b> | <b>0.310</b> | <b>268.8</b> | <b>1.93</b> | <b>4.59</b> | <b>0.80</b> | 0.60        |
| DINOv3-B       | 100K  | 0.749        | 63.64        | 21.4        | 6.14        | 0.63        | 0.60        | 0.571        | 144.0        | 7.57        | 6.09        | 0.76        | 0.53        |
| <b>+iREPA</b>  | 100K  | <b>0.651</b> | <b>78.79</b> | <b>16.2</b> | <b>6.14</b> | <b>0.66</b> | <b>0.61</b> | <b>0.481</b> | <b>181.9</b> | <b>4.87</b> | 6.10        | <b>0.78</b> | <b>0.55</b> |
| DINOv3-B       | 400K  | 0.474        | 126.7        | 8.10        | 5.06        | 0.70        | 0.66        | 0.336        | 261.2        | 1.99        | 4.58        | 0.80        | 0.61        |
| <b>+iREPA</b>  | 400K  | <b>0.441</b> | <b>132.9</b> | <b>7.13</b> | <b>4.93</b> | <b>0.71</b> | <b>0.66</b> | <b>0.314</b> | <b>272.4</b> | <b>1.89</b> | <b>4.58</b> | <b>0.80</b> | 0.60        |
| WebSSL-1B      | 100K  | 0.825        | 53.87        | 25.5        | 6.57        | 0.61        | 0.59        | 0.622        | 124.0        | 9.59        | 6.37        | 0.76        | 0.52        |
| <b>+iREPA</b>  | 100K  | <b>0.653</b> | <b>77.47</b> | <b>16.6</b> | <b>6.18</b> | <b>0.66</b> | <b>0.61</b> | <b>0.484</b> | <b>177.6</b> | <b>5.09</b> | <b>6.11</b> | <b>0.79</b> | <b>0.54</b> |
| WebSSL-1B      | 400K  | 0.512        | 116.9        | 9.39        | 5.14        | 0.70        | 0.64        | 0.354        | 250.5        | 2.24        | 4.61        | 0.81        | 0.58        |
| <b>+iREPA</b>  | 400K  | <b>0.445</b> | <b>130.8</b> | <b>7.48</b> | <b>4.91</b> | <b>0.70</b> | <b>0.65</b> | <b>0.311</b> | <b>271.6</b> | <b>1.90</b> | <b>4.58</b> | 0.80        | <b>0.61</b> |
| PE-Core-G      | 100K  | 0.922        | 42.74        | 32.3        | 6.70        | 0.57        | 0.59        | 0.714        | 97.2         | 14.1        | 6.56        | 0.71        | 0.53        |
| <b>+iREPA</b>  | 100K  | <b>0.697</b> | <b>75.01</b> | <b>18.1</b> | <b>6.03</b> | <b>0.64</b> | <b>0.61</b> | <b>0.525</b> | <b>176.8</b> | <b>5.66</b> | <b>6.08</b> | <b>0.77</b> | <b>0.54</b> |
| PE-Core-G      | 400K  | 0.525        | 109.4        | 10.4        | 5.00        | 0.69        | 0.64        | 0.366        | 238.4        | 2.44        | 4.57        | 0.81        | 0.59        |
| <b>+iREPA</b>  | 400K  | <b>0.458</b> | <b>132.0</b> | <b>7.78</b> | 5.02        | <b>0.70</b> | <b>0.65</b> | <b>0.322</b> | <b>275.4</b> | <b>1.93</b> | 4.59        | 0.80        | <b>0.61</b> |
| CLIP-L         | 100K  | 0.790        | 54.07        | 25.2        | 6.65        | 0.61        | 0.60        | 0.605        | 124.5        | 9.77        | 6.59        | 0.75        | 0.52        |
| <b>+iREPA</b>  | 100K  | <b>0.657</b> | <b>74.46</b> | <b>17.8</b> | <b>6.33</b> | <b>0.65</b> | <b>0.61</b> | <b>0.489</b> | <b>172.0</b> | <b>5.67</b> | <b>6.33</b> | <b>0.78</b> | <b>0.54</b> |
| CLIP-L         | 400K  | 0.487        | 117.8        | 8.97        | 4.98        | 0.70        | 0.65        | 0.339        | 258.7        | 2.15        | 4.64        | 0.81        | 0.59        |
| <b>+iREPA</b>  | 400K  | <b>0.442</b> | <b>130.1</b> | <b>7.42</b> | 5.03        | <b>0.71</b> | <b>0.65</b> | <b>0.307</b> | <b>271.8</b> | <b>1.96</b> | <b>4.64</b> | 0.80        | <b>0.61</b> |

Table 8: **Additional results with alternative evaluation metrics.** We report additional results under the CMMD metric (Jayasumana et al., 2024). All results use SiT-XL/2, batch size 256, and traditional REPA as the baseline; CMMD is computed with the PyTorch implementation of (Paul, 2024), using the standard reference set from the OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021) as reference images. Across every vision encoder, accentuating the transfer of spatial features (iREPA) improves convergence speed under both CMMD and the traditional metrics (IS, FID, etc.), at both 100K and 400K steps and both with and without CFG—without a single exception. The key findings of §4 are therefore robust to the choice of evaluation metric rather than an artifact of FID.

## M.6 SPATIAL IMPROVEMENTS WITH SAM2

Table 9 and Figure 30 study the impact of spatial normalization (§4) on SAM2-S (46M) encoder features<sup>2</sup>. As Table 9 shows, spatial normalization does improve SAM2 performance, albeit by a modest margin. The reason is instructive and fully consistent with our account: spatial normalization operates by removing the global component (the mean of the patch tokens) to enhance spatial contrast, and because SAM2 already carries little to no global information (validation accuracy < 24%), there is scarcely any global component left to remove, so the spatial contrast is only slightly sharpened (see Figure 30). The modest gain is therefore predicted by, rather than at odds with, the spatial-structure account.

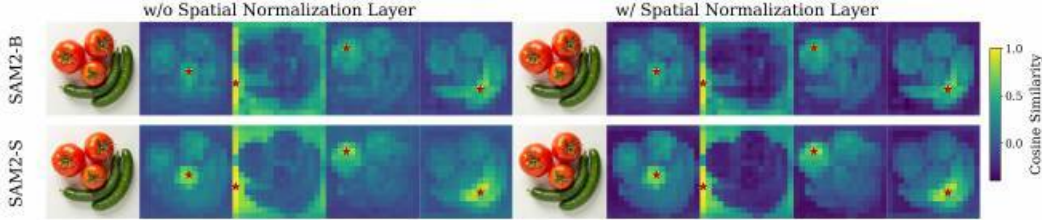


Figure 30: **Impact of spatial normalization on SAM2.** For SAM2, the spatial normalization layer does enhance spatial contrast, but only by a modest margin. The reason is consistent with our account: spatial normalization (§4) operates by removing the global component (the mean of the patch tokens), and because SAM2 already carries little to no global information (validation accuracy < 24%), there is little to remove, so the spatial contrast is only slightly sharpened.

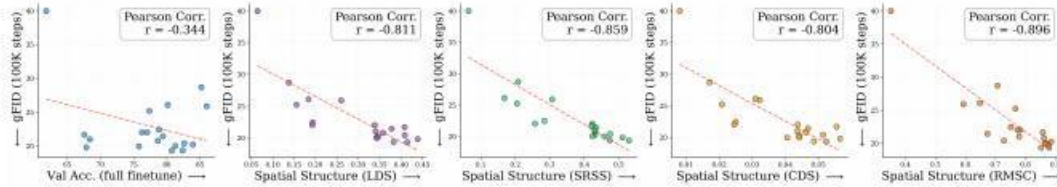
| Vision Encoder | Steps | IS $\uparrow$ | FID $\downarrow$ | sFID $\downarrow$ | Prec. $\uparrow$ | Rec. $\uparrow$ |
|----------------|-------|---------------|------------------|-------------------|------------------|-----------------|
| SAM2-S         | 100K  | 50.69         | 25.32            | 6.52              | 0.631            | 0.588           |
| <b>+iREPA</b>  | 100K  | <b>53.14</b>  | <b>24.52</b>     | <b>6.28</b>       | <b>0.629</b>     | <b>0.591</b>    |
| SAM2-S         | 400K  | 110.77        | 9.54             | 4.93              | 0.699            | 0.638           |
| <b>+iREPA</b>  | 400K  | <b>114.62</b> | <b>9.10</b>      | <b>4.89</b>       | <b>0.704</b>     | <b>0.640</b>    |

Table 9: **Impact of spatial improvements on SAM2.** iREPA improves generation over baseline REPA at both 100K and 400K steps; the margin is modest precisely because SAM2 already carries little global information for spatial normalization to remove—so the modest gain is itself predicted by the spatial-structure account. All results use SiT-XL/2, without classifier-free guidance, SAM2-S (46M) as the vision encoder, and traditional REPA as the baseline.

<sup>2</sup>Note that similar to (Bolya et al., 2025), we use the intermediate output of vision encoder for SAM2 features and not the mask logits. As shown in (Bolya et al., 2025), while mask logits lead to sharper spatial maps, the mask logits itself are not suitable as a target representation.

## M.7 ADDITIONAL RESULTS WITH FULL-FINETUNING ACCURACY INSTEAD OF LINEAR PROBING ACCURACY

**Correlation analysis with full-finetuning accuracy instead of linear probing accuracy.** We repeat the correlation analysis of §3 using validation accuracy after full-finetuning<sup>3</sup> in place of linear probing accuracy. Results are reported in Fig. 31. All results use SiT-XL/2 (100K steps) and REPA; for full-finetuning validation accuracy, we finetune the entire vision encoder with a learning rate of  $5e-5$ , a warmup ratio of 0.1, and 30 epochs in total. Every spatial metric again correlates far more strongly with generation performance than full-finetuning accuracy does. Strikingly, gFID exhibits a weak *positive* correlation with full-finetuning validation accuracy (Pearson  $r = 0.317$ )—as validation accuracy rises, gFID rises and generation deteriorates. This reinforces the observations of §2, in which target representations with higher global semantic performance frequently yield similar or worse generation under representation alignment (REPA), and confirms that the conclusion is unchanged whether global information is measured by linear probing or by full finetuning.



**Figure 31: Correlation analysis with full-finetuning accuracy instead of linear probing accuracy.** We repeat the correlation analysis of §3 using validation accuracy after full-finetuning in place of linear probing accuracy. All results use SiT-XL/2 (100K steps) and REPA. Every spatial metric again correlates far more strongly with generation performance than full-finetuning accuracy does. Strikingly, gFID exhibits a weak positive correlation with full-finetuning validation accuracy (Pearson  $r = 0.317$ )—as validation accuracy rises, gFID rises and generation deteriorates. This reinforces the observations of §2, in which target representations with higher global semantic performance frequently yield similar or worse generation under representation alignment (REPA).

<sup>3</sup>Please note that while the final findings remain similar, in context of REPA linear probing might be more accurate for estimating global information in “pretrained” vision encoders. This is because REPA uses the “pretrained” encoder features themselves for regularization. Finetuning the encoder itself can impact the amount of global information. Same as REPA (Yu et al., 2024), we therefore use linear probing accuracy as default for measuring global information.

## M.8 ADDITIONAL RESULTS ON PIXEL-SPACE DIFFUSION MODELS

| Method                                 | #Params | IS $\uparrow$ | FID $\downarrow$ | sFID $\downarrow$ | Prec. $\uparrow$ | Rec. $\uparrow$ |
|----------------------------------------|---------|---------------|------------------|-------------------|------------------|-----------------|
| PixelFlow (Chen et al., 2025)          | 459M    | 24.67         | 54.33            | 9.71              | -                | 0.58            |
| PixDDT (Wang et al., 2025b)            | 434M    | 36.24         | 46.37            | 17.14             | -                | 0.63            |
| PixNerd (Wang et al., 2025a)           | 458M    | 43.01         | 37.49            | 10.65             | -                | 0.62            |
| DeCo w/o $\mathcal{L}_{\text{FreqFM}}$ | 426M    | 46.44         | 34.12            | 10.41             | -                | 0.64            |
| DeCo + REPA (Ma et al., 2025b)         | 426M    | 48.35         | 31.35            | 9.34              | -                | 0.65            |
| JiT (Li & He, 2025a)                   | 459M    | 29.37         | 49.06            | 11.21             | 0.40             | 0.62            |
| <b>JiT+iREPA (Ours)</b>                | 459M    | <b>50.72</b>  | <b>29.19</b>     | 9.42              | <b>0.51</b>      | <b>0.65</b>     |

Table 10: **Generation performance of pixel-space diffusion models.** Despite its simplicity, iREPA combined with JiT (Li & He, 2025a) outperforms recently proposed state-of-the-art pixel-space diffusion methods, DeCo (Ma et al., 2025b) among them, attaining the best FID and IS in the table. All results are reported at 200K training iterations with a batch size of 256, evaluated using 50-step Euler sampling without classifier-free guidance; results for DeCo are taken directly from (Ma et al., 2025b).